Unit

40 Factors affecting chemical equilibrium systems

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

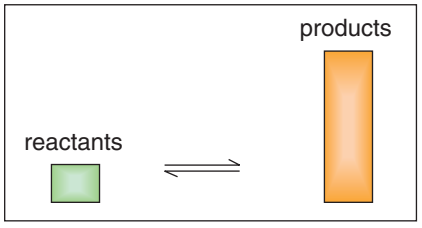
- Le Chatelier's principle
- The effect of concentration changes, pressure changes and temperature changes on chemical equilibrium systems
- Applying principles of equilibria and reaction rates to the Haber process

40.1 Position of equilibrium

The proportion of products to reactants in an equilibrium mixture is described as the **position of equilibrium**.

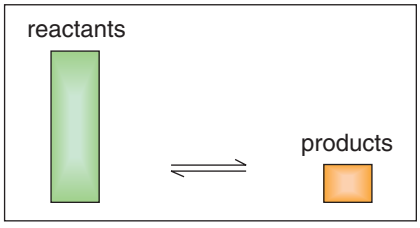
If the product concentrations at equilibrium are high and hardly any reactants are left, we say the position of equilibrium lies to the product side (Fig. 40.1). The value of K_c is large.

If the reactant concentrations at equilibrium are high and product concentrations are very low, we say the position of equilibrium lies to the reactant side (Fig. 40.2). The value of K_c is small.



system at equilibrium

Fig. 40.1 The position of this equilibrium lies to the product side



system at equilibrium

Fig. 40.2 The position of this equilibrium lies to the reactant side

40.2 Effect of changing conditions on systems in equilibrium

To achieve the best yields from industrial processes involving reversible reactions, chemists need to understand the effect of changing the conditions on the position of equilibrium.

If the proportion of products in the equilibrium mixture has increased, we say that 'the position of equilibrium has shifted to the right'. If the proportion of reactants in the equilibrium mixture has increased, we say that 'the position of equilibrium has shifted to the left'.

position of equilibrium 平衡位置



Fig. 40.3 Henri Le Chatelier

The position of equilibrium can be altered by changing the following conditions in a system:

- concentrations of the reactants or products;
- pressure in reactions involving gases; and
- temperature.

The effect of a change can be predicted using **Le Chatelier's principle**, formulated by a French chemist Henri Le Chatelier (Fig. 40.3) in 1884.

✓ Le Chatelier's principle states that if the condition of a system in equilibrium is changed, the position of equilibrium will shift so as to reduce that change.

40.3 The effect of concentration changes on chemical equilibrium systems

Effect of concentration changes on the equilibrium system established in aqueous solution of potassium chromate

Reference Website
Visit the following website to explore the effect of changing the amounts of reactants and products on the equilibrium composition of a system:
http://www.chm.davidson.edu/vce/equilibria/LeChatelier.html
(accessed April 2015)

Video
Demonstrating the effect of adding acid and alkali on a chemical equilibrium system containing $Cr_2O_7^{2-}$ ions and CrO_4^- ions

Consider the equilibrium system attained in an aqueous solution of potassium chromate:

$$\underset{\text{yellow}}{2CrO_4^{2-}(aq)} + 2H^+(aq) \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + H_2O(l)$$

We can explain the colour changes of the solution upon the addition of dilute sulphuric acid or dilute sodium hydroxide solution by Le Chatelier's principle.

Le Chatelier's principle 勒沙得利爾原理

Effect of adding dilute sulphuric acid

When a small amount of dilute sulphuric acid is added to the system, the colour of the solution changes gradually from yellow to orange. This indicates that the concentration of $CrO_4^{2-}(aq)$ ions has decreased while that of $Cr_2O_7^{2-}(aq)$ ions has increased (Fig. 40.4).

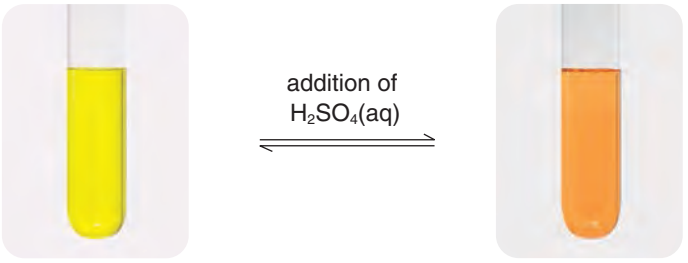


Fig. 40.4 Effect of adding dilute sulphuric acid to the equilibrium system established in an aqueous solution of potassium chromate

Upon the addition of the acid, the forward reaction goes faster while the backward reaction goes on more or less at the same rate as before.

When hydrogen ions are added, the system responds by reducing this change. A net forward reaction occurs to consume some of the hydrogen ions. The position of equilibrium shifts to the right.

Effect of adding dilute sodium hydroxide solution

When a small amount of dilute sodium hydroxide solution is added to the resulting solution, the yellow colour is restored. This indicates that the concentration of $CrO_4^{2-}(aq)$ ions has increased while that of $Cr_2O_7^{2-}(aq)$ ions has decreased (Fig. 40.5).

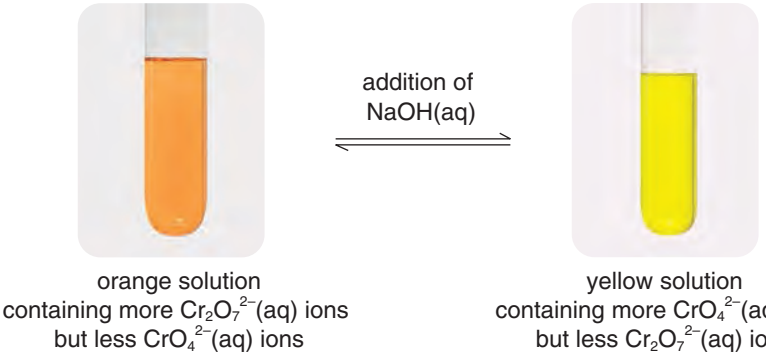


Fig. 40.5 Effect of adding dilute sodium hydroxide solution to the resulting solution

When the alkali is added, the hydroxide ions react with hydrogen ions to form water. Thus, the concentration of hydrogen ions decreases. The system responds by reducing this change. A net backward reaction occurs to produce more hydrogen ions. The position of equilibrium shifts to the left.

Upon the addition of the alkali, the backward reaction goes faster while the forward reaction goes on more or less at the same rate as before.

Table 40.1 summarizes the effect of adding dilute sulphuric acid or dilute sodium hydroxide solution to the equilibrium system.

Table 40.1

A summary of effect of adding H_2SO_4(aq) or NaOH(aq) to the equilibrium system
$2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$

Action on equilibrium system	Colour change observed	System responds by	Direction of net reaction	Shift of equilibrium position
Adding $H^+(aq)$	from yellow to orange i.e. $[CrO_4^{2-}(aq)]$ decreases $[Cr_2O_7^{2-}(aq)]$ increases	consuming some $H^+(aq)$ ions	a net forward reaction occurs	position of equilibrium shifts to the right $2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$ yellow → orange
Adding $OH^-(aq)$	from orange to yellow i.e. $[CrO_4^{2-}(aq)]$ increases $[Cr_2O_7^{2-}(aq)]$ decreases	producing more $H^+(aq)$ ions	a net backward reaction occurs	position of equilibrium shifts to the left $2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$ yellow ← orange

Results of the experiment can be summarized by the following statement:

✓ If the concentration of one of the species in an equilibrium mixture is changed, the position of equilibrium will shift so as to reduce the change in concentration. N1

N1 Refer to page T24.

40.1
Investigating the effect of concentration changes on equilibrium.

Video
Investigating the effect of concentration changes on equilibrium

The statement above is a specific application of Le Chatelier's principle.

One important point is that although the concentration of individual species in the equilibrium mixture may vary, the value of equilibrium constant, K_c, is always the same at one particular temperature.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

When bromine is dissolved in water, the following equilibrium system is established:

$$\underset{\text{yellow-brown}}{Br_2(l)} + H_2O(l) \rightleftharpoons \underbrace{H^+(aq) + Br^-(aq) + HOBr(aq)}_{\text{colourless}}$$

When a little dilute sodium hydroxide solution is added to the system, a colourless solution results. When a little dilute sulphuric acid is added to the resulting solution, the yellow-brown colour of the solution is restored (Fig. 40.6).

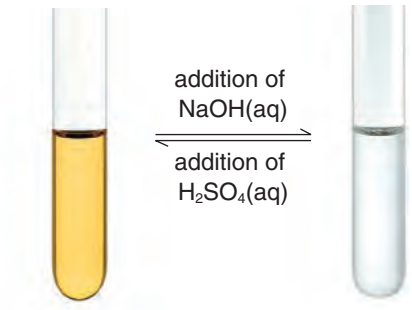


Fig. 40.6 The experimental result

Explain the observations using Le Chatelier's principle.
Refer to page T24.

Do you know

An important biological application of equilibrium — transport of oxygen by the blood

As blood passes through the lungs where the concentration of oxygen is high, haemoglobin in red blood cells combines with oxygen to form oxyhaemoglobin.

haemoglobin + oxygen $\rightleftharpoons$ oxyhaemoglobin

This is a reversible reaction. The concentration of oxygen is much lower in the body tissue. When the blood reaches the body tissue, the position of equilibrium of the reaction shifts to the left. Oxygen is released to the tissue.

Effect of concentration changes on an equilibrium system of dinitrogen tetroxide and nitrogen dioxide

Simulation
Effect of concentration changes on chemical equilibrium system

Dinitrogen tetroxide (N_2O_4) decomposes to form nitrogen dioxide (NO_2) according to the equation:

$$\underset{\text{pale yellow}}{N_2O_4(g)} \rightleftharpoons \underset{\text{dark brown}}{2NO_2(g)}$$

ASL 2006 Q4(b)

Effect of adding extra dinitrogen tetroxide (N_2O_4)

Consider an equilibrium system of $N_2O_4(g)$ and $NO_2(g)$. Suppose suddenly extra $N_2O_4(g)$ is added to the system while the volume and the temperature are both kept constant. The mixture gets paler for a moment and then becomes darker brown (Fig. 40.7a–c).

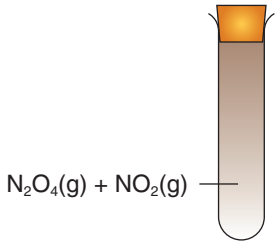


Fig. 40.7a $N_2O_4(g)$ in equilibrium with $NO_2(g)$

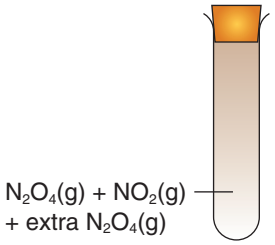


Fig. 40.7b Suddenly extra pale yellow $N_2O_4(g)$ is added

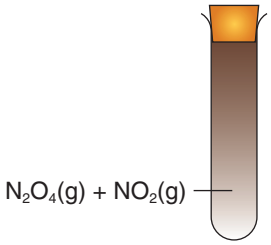


Fig. 40.7c The position of equilibrium shifts to the right, more dark brown $NO_2(g)$ is present

The mixture gets paler first due to the addition of pale yellow $N_2O_4(g)$. According to Le Chatelier's principle, the system responds by reducing this change. A net forward reaction occurs to convert more $N_2O_4(g)$ to $NO_2(g)$. This results in an increase in the concentration of the dark brown $NO_2(g)$ and a decrease in the concentration of the pale yellow $N_2O_4(g)$. A darker brown colour is thus observed.

We can say that 'the position of equilibrium shifts to the right'.

adding extra $N_2O_4(g)$,
position of equilibrium shifts to the right

$$\underset{\text{pale yellow}}{N_2O_4(g)} \rightleftharpoons \underset{\text{dark brown}}{2NO_2(g)}$$

Fig. 40.8 shows how concentrations of substances in the system change with time after extra $N_2O_4(g)$ is added. Notice that a new state of equilibrium is attained after some time.

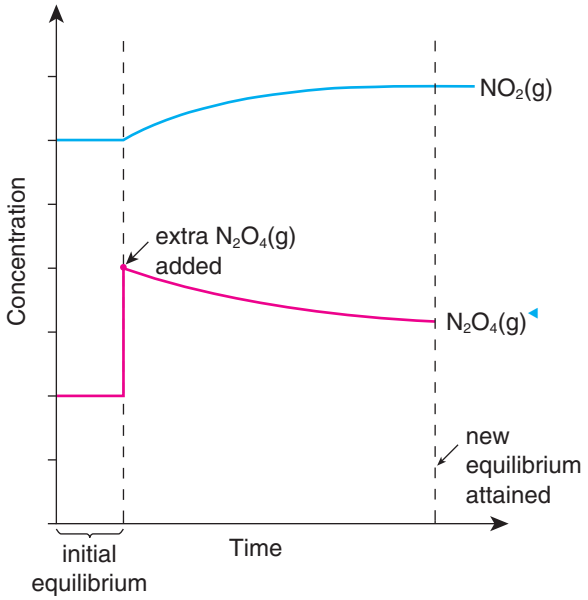


Fig. 40.8 A concentration-time graph for the system of $N_2O_4(g)$ and $NO_2(g)$

The final concentration of $N_2O_4(g)$ is between the maximum value when the extra $N_2O_4(g)$ is suddenly added and the value at the initial equilibrium.

Summary of effect of concentration changes on a chemical equilibrium system

Table 40.2 lists the effect of changing the concentration of a reactant or product on the general chemical equilibrium system:

$$A + B \rightleftharpoons C + D$$

Table 40.2

A summary of effect of changing the concentration of a reactant or product on the general chemical equilibrium system $A + B \rightleftharpoons C + D$

Action on equilibrium system	System responds by	The result	Direction of net reaction	Shift of position of equilibrium
Increasing the concentration of A	consuming some of A via reaction with B	more C and D form	a net forward reaction occurs	to the right
Decreasing the concentration of A	producing more A via the reaction of C and D	more B but less C and D in the new equilibrium mixture	a net backward reaction occurs	to the left
Increasing the concentration of C	consuming some of C via reaction with D	more A and B form	a net backward reaction occurs	to the left
Decreasing the concentration of C	producing more C via the reaction of A and B	more D but less A and B in the new equilibrium mixture	a net forward reaction occurs	to the right

Example 40.1

Q Methane reacts with steam according to the following equation:

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$$

The following changes are made to an equilibrium mixture of the four gases. In each case, predict and explain what will happen to the yield of hydrogen.

a) The concentration of $H_2O(g)$ is increased.

b) The concentration of $CH_4(g)$ is decreased.

A a) When the concentration of $H_2O(g)$ is increased, the system responds by reducing this change.

A net forward reaction occurs to consume some of the $H_2O(g)$. The position of equilibrium shifts to the right. Thus, the yield of $H_2(g)$ increases.

b) When the concentration of $CH_4(g)$ is decreased, the system responds by reducing this change.

A net backward reaction occurs to produce more $CH_4(g)$. The position of equilibrium shifts to the left. Thus, the yield of $H_2(g)$ decreases.

Practice 40.1

1 Consider the equilibrium system:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Give the effect of each change listed in the table below on the system.

(The volume and the temperature are both kept constant.)

Change on system	System responds by	Direction of net reaction	Shift of position of equilibrium
a) Adding some $PCl_5(g)$	consuming some $PCl_5(g)$	a net forward reaction occurs	to the right
b) Removing some $PCl_5(g)$	producing more $PCl_5(g)$	a net backward reaction occurs	to the left
c) Adding some $Cl_2(g)$	consuming some $Cl_2(g)$	a net backward reaction occurs	to the left
d) Removing some $PCl_3(g)$	producing more $PCl_3(g)$	a net forward reaction occurs	to the right

Continued on next page

2 $Fe^{3+}(aq)$ ions and $SCN^-(aq)$ ions react in solution to give deep red $Fe(SCN)^{2+}(aq)$ ions.

$$\underset{\text{yellow-brown}}{Fe^{3+}(aq)} + \underset{\text{colourless}}{SCN^-(aq)} \rightleftharpoons \underset{\text{deep red}}{Fe(SCN)^{2+}(aq)}$$



$Fe^{3+}(aq)$ ions



$SCN^-(aq)$ ions



$Fe(SCN)^{2+}(aq)$ ions

A mixture of the three ions is allowed to attain equilibrium. Give your expected observation in each of the following cases. Explain in terms of Le Chatelier's principle.

a) KSCN(aq) is added to the equilibrium mixture.

b) $Na_3PO_4(s)$ is added to the equilibrium mixture (it is known that $FePO_4(s)$ is insoluble in water).

DSE 2012 Paper 1B Q13(b)

3 Consider the equilibrium system:

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

Predict the effect of increasing the concentration of NO(g) on

a) the rate of formation of $NO_2(g)$;

b) the position of equilibrium;
The position of equilibrium would shift to the right.

c) the value of K_c.
The value of K_c would remain unchanged.

(The volume and temperature are both kept constant.)

3 a) Increasing the concentration of NO(g) would increase the rate of the forward reaction, i.e. increase the rate of formation of $NO_2(g)$.

Practice 40.1

2 a) The deep red colour of the mixture becomes more intense.
When KSCN(aq) is added, the concentration of $SCN^-(aq)$ ions increases. The system responds by reducing this change. A net forward reaction occurs to consume some of the $SCN^-(aq)$ ions.

b) The deep red colour of the mixture becomes less intense.
When $Na_3PO_4(s)$ is added, the $PO_4^{3-}(aq)$ ions react with $Fe^{3+}(aq)$ ions to form $FePO_4(s)$. The concentration of $Fe^{3+}(aq)$ ions decreases. The system responds by reducing this change. A net backward reaction occurs to produce more $Fe^{3+}(aq)$ ions.

40.4 Predicting the shift in equilibrium position using the reaction quotient, Q_c

Consider an equilibrium system of $N_2O_4(g)$ and $NO_2(g)$ kept at a constant temperature.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

The equilibrium concentrations are as follows:

$$[N_2O_4(g)] = 0.200 \text{ mol dm}^{-3}$$

$$[NO_2(g)] = 1.60 \text{ mol dm}^{-3}$$

$$\therefore \quad K_c = \frac{[NO_2(g)]^2}{[N_2O_4(g)]} = \frac{(1.60 \text{ mol dm}^{-3})^2}{0.200 \text{ mol dm}^{-3}} = 12.8 \text{ mol dm}^{-3}$$

Suppose the concentration of $N_2O_4(g)$ is now increased to $0.300 \text{ mol dm}^{-3}$. The reaction quotient, Q_c, of the system is as follows:

$$Q_c = \frac{[NO_2(g)]^2}{[N_2O_4(g)]} = \frac{(1.60 \text{ mol dm}^{-3})^2}{0.300 \text{ mol dm}^{-3}} = 8.53 \text{ mol dm}^{-3}$$

Q_c is less than K_c. In order to attain a new state of equilibrium, Q_c must increase, i.e. the concentration of $NO_2(g)$ must increase while that of $N_2O_4(g)$ must decrease. Thus, a net forward reaction occurs, i.e. the position of equilibrium shifts to the right — as predicted by Le Chatelier's principle. N2

N2 Refer to page T25.

The concept of reaction quotient makes it possible to explain the effect of changing the concentraion of one of the species in an equilibrium mixture quantitatively.

40.5 Effect of pressure changes on chemical equilibrium systems

Generally speaking, pressure has little effect on the chemistry of solids and liquids. Pressure does not affect the concentrations of solids and liquids — particles in solids and liquids are already in contact and it is difficult to push them closer together.

Pressure does have significant effects on the chemistry of reacting gases. As shown in Fig. 40.9, the concentration of a gas increases with an increase in pressure and decreases with a decrease in pressure. Since chemical equilibrium systems are influenced by concentration changes, pressure changes also have an effect on equilibrium systems where a gas is involved.

Reference Website
Visit the following website to explore the effect of a change in volume on the equilibrium composition of a system:
http://www.chm.davidson.edu/vce/equilibria/Volume.html
(accessed April 2015)

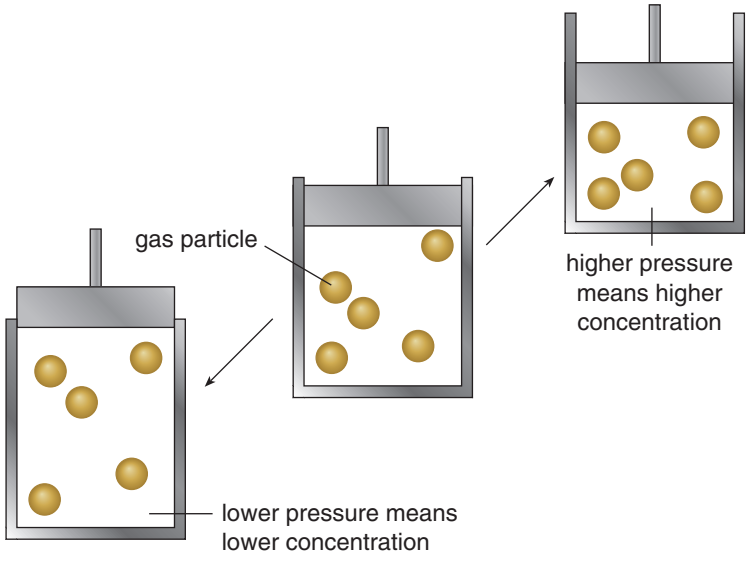


Fig. 40.9 Pressure has a considerable effect on the concentration of a gas

Effect of pressure change on an equilibrium system of nitrogen, hydrogen and ammonia

Fig. 40.10a shows a container holding a mixture of nitrogen, hydrogen and ammonia at equilibrium. The gases react according to the following equation:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

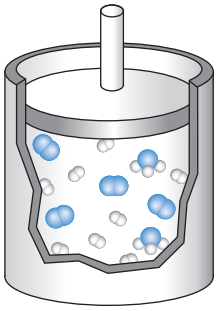
The equation has 4 moles of gas on the left-hand side and 2 moles of gas on the right-hand side.

Suppose we reduce the volume of the container by one-half suddenly (Fig. 40.10b). This doubles the total pressure of gases in the container.

According to Le Chatelier's principle, the system responds by reducing the pressure. This brings about a net reaction that decreases the number of moles of gas as fewer gas molecules exert less pressure. A net forward reaction occurs because the forward reaction converts 4 moles of gas to 2 moles of gas (Fig. 40.10c). The position of equilibrium shifts to the right, the side with a fewer number of moles of gas. N3

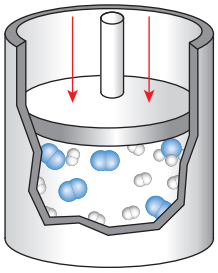
The pressure of a gas arises from gas molecules hitting the walls of the container. Fewer collisions (per unit area per unit time) occur between molecules and walls when there are fewer gas molecules.

N3 Refer to page T26.



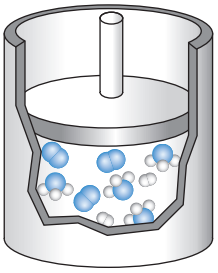
(a)

Fig. 40.10a A mixture of nitrogen, hydrogen and ammonia at equilibrium



(b)

Fig. 40.10b The total pressure of gases in the container is doubled



(c)

Fig. 40.10c A net forward reaction occurs

key: $N_2(g)$ $H_2(g)$ $NH_3(g)$

✓ Pressure increase:

position of equilibrium shifts to the side with a fewer number of moles of gas

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Effect of pressure change on an equilibrium system of dinitrogen tetroxide and nitrogen dioxide

Simulation
Effect of pressure changes on chemical equilibrium system

Consider the equilibrium system of dinitrogen tetroxide (N_2O_4) and nitrogen dioxide (NO_2):

$$\underset{\text{pale yellow}}{N_2O_4(g)} \rightleftharpoons \underset{\text{dark brown}}{2NO_2(g)}$$

The equation has 1 mole of gas on the left-hand side and 2 moles of gas on the right-hand side.

Fig. 40.11a shows a gas syringe containing a pale brown mixture of $N_2O_4(g)$ and $NO_2(g)$ at equilibrium. Suppose we suddenly decrease the pressure of the system by pulling out the plunger (Fig. 40.11b). The mixture gets paler for a moment and then becomes darker brown gradually (Fig. 40.11c).

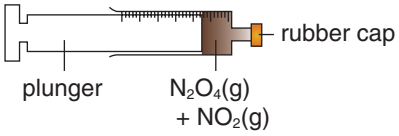


Fig. 40.11a A mixture of $N_2O_4(g)$ and $NO_2(g)$ at equilibrium

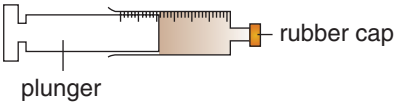


Fig. 40.11b The pressure of the system is suddenly decreased

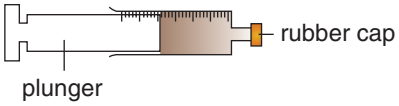


Fig. 40.11c The mixture gets paler for a moment and then becomes darker brown gradually

We have increased the volume of the mixture and thus decreased the concentration of the dark brown $NO_2(g)$. Hence the mixture gets paler for a moment.

According to Le Chatelier's principle, the system responds by increasing the pressure. This brings about a net reaction that increases the number of moles of gas as more gas molecules exert more pressure. A net forward reaction occurs because the forward reaction converts 1 mole of $N_2O_4(g)$ to 2 moles of $NO_2(g)$. The position of equilibrium shifts to the right, the side with a greater number of moles of gas.

✓ Pressure decrease:

position of equilibrium shifts to
the side with a greater number of moles of gas

$$\xrightarrow{\qquad} \quad N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

Fig. 40.12 shows how the concentration of $NO_2(g)$ in the system changes with time.

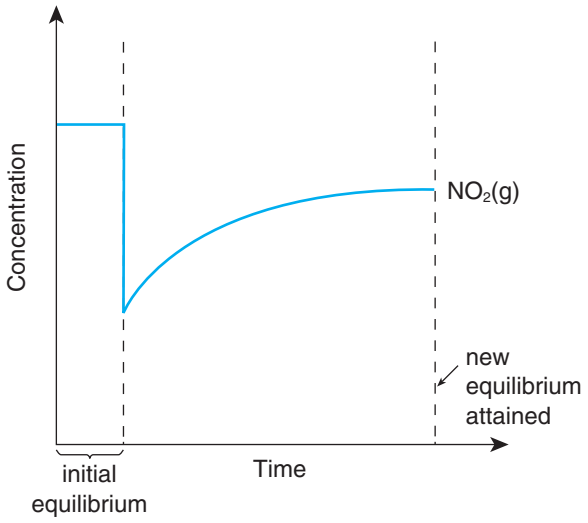


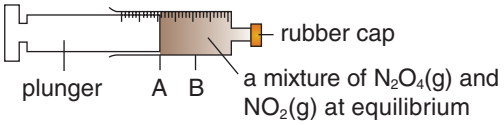
Fig. 40.12 The concentration-time graph of $NO_2(g)$ for the system of $N_2O_4(g)$ and $NO_2(g)$

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

The diagram below shows a gas syringe containing a pale brown mixture of $N_2O_4(g)$ and $NO_2(g)$ at equilibrium.

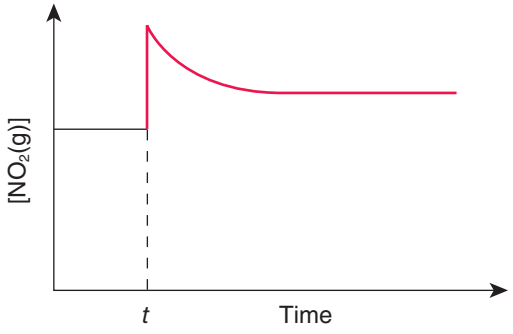
$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$
pale yellow — dark brown



The plunger is quickly pushed from position A to position B at time *t*.

a) State and explain the expected observations.

b) Sketch on the given graph to show the expected variation in the concentration of $NO_2(g)$ in the mixture until a new equilibrium is attained.



a) The brown colour of the mixture becomes more intense for a moment and then gets paler gradually.
When the volume of the syringe is decreased, the concentration of $NO_2(g)$ and the pressure of the system increase.
According to Le Chatelier's principle, the system responds by reducing the pressure. This brings about a net reaction that decreases the number of moles of gas.
A net backward reaction occurs because the backward reaction converts 2 moles of $NO_2(g)$ to 1 mole of $N_2O_4(g)$.
The position of equilibrium shifts to the left. So $[NO_2(g)]$ decreases.

ASL 2006 Q4(a)

Effect of pressure change on an equilibrium system of hydrogen, iodine and hydrogen iodide

Consider the following equilibrium system:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

The number of moles of gas is the same on both sides, so changing the pressure has no effect on the position of equilibrium.

ASL 2008 Paper 2 Q4(a)(ii)(I)

Summary of effect of pressure changes on chemical equilibrium systems

In general, Le Chatelier's principle predicts that

- a pressure increase (a volume decrease) will bring about a net reaction that decreases the number of moles of gas, i.e. the position of equilibrium will shift to the side of the equation with a fewer number of moles of gas.

- a pressure decrease (a volume increase) will bring about a net reaction that increases the number of moles of gas, i.e. the position of equilibrium will shift to the side of the equation with a greater number of moles of gas.

AL 2006 Paper 2 Q4(c)(ii)

Table 40.3 summarizes the effect of pressure changes on a chemical equilibrium system.

An increase in pressure (decrease in volume) increases concentrations of species in the system. Thus, the rates of both the forward reaction and backward reaction increase.

Table 40.3

A summary of effect of pressure changes on an equilibrium system
$aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$

Pressure (volume) change on equilibrium system		Rates of forward and backward reactions	Direction of net reaction	Shift of position of equilibrium
Pressure increase (volume decrease)	a + b > c + d	rates of both reactions increase, but the forward reaction rate increases more	a net forward reaction occurs	to the right
	a + b = c + d	same increase in rates of both reactions	no net reaction	no change
	a + b < c + d	rates of both reactions increase, but the backward reaction rate increases more	a net backward reaction occurs	to the left

Continued on next page

Pressure (volume) change on equilibrium system		Rates of forward and backward reactions	Direction of net reaction	Shift of position of equilibrium
Pressure decrease (volume increase)	a + b > c + d	rates of both reactions decrease, but the forward reaction rate decreases more	a net backward reaction occurs	to the left AL 2009 Paper 2 Q2(b)(ii)
	a + b = c + d	same decrease in rates of both reactions	no net reaction	no change
	a + b < c + d	rates of both reactions decrease, but the backward reaction rate decreases more	a net forward reaction occurs	to the right

N4 Refer to page T27.

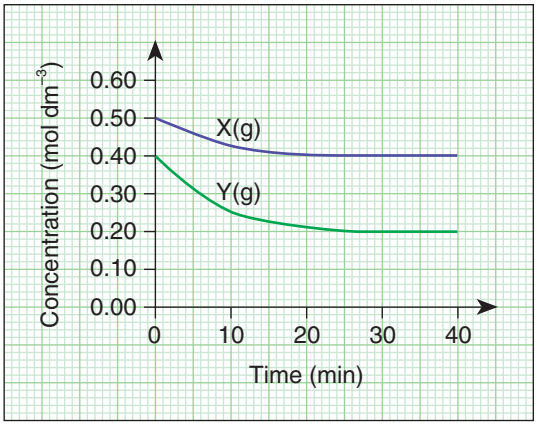
N4

Example 40.2

Q X(g) reacts with Y(g) reversibly to give Z(g).

$$X(g) + 2Y(g) \rightleftharpoons 2Z(g)$$

A mixture of X(g) and Y(g) is allowed to react in a closed container of volume 1 dm^3 kept at a constant temperature. The graph below shows the changes in concentrations of X(g) and Y(g) in the container with time.



a) Sketch, in the same graph, the variation of concentration of Z(g) with time.

b) Suppose the mixture of X(g) and Y(g) is allowed to react at the same temperature but in a closed container of volume 2 dm^3 instead.

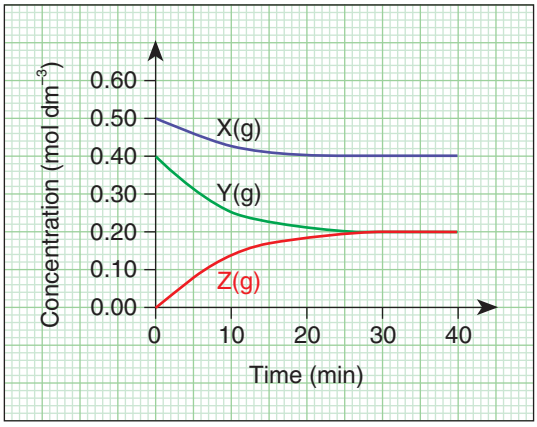
i) What will happen to the rate of the forward reaction? Explain your answer.

ii) How will the yield of Z(g) be affected? Explain your answer.

ASL 2005 Q8(b)(ii)(1)

Continued on next page

A1 a)



b) i) The rate of the forward reaction will decrease.

Increasing the volume of the container causes the concentrations of reactants to decrease.

The chance of collision between reactant particles decreases and the frequency of effective collision decreases.

ii) The yield of Z(g) will decrease.

The pressure of the system decreases when the volume of the closed container is increased. The position of equilibrium will shift to the side with a greater number of moles of gas (i.e. to the left).

Practice 40.2

1 In which of the following reactions will the position of equilibrium be affected by changing the pressure? Explain your answers.

a) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ Yes
This reaction involves gases and there are different number of moles of gas on the two sides of the equation.

b) $CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$ No
This reaction does not involve any gas.

c) $H_2(g) + CO_2(g) \rightleftharpoons H_2O(g) + CO(g)$ No
This reaction involves gas but the number of moles of gas is the same on both sides of the equation.

2 Consider the following reversible reaction:

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

A mixture of the three gases was allowed to attain equilibrium in a gas syringe.

Give the effect of each pressure change on items listed in the table below.

Change on system	System responds by	Direction of net reaction	Shift of position of equilibrium
Increasing the pressure	reducing the pressure (i.e. decreasing the number of moles of gas)	a net forward reaction occurs	to the right
Decreasing the pressure	increasing the pressure (i.e. increasing the number of moles of gas)	a net backward reaction occurs	to the left

40.6 Effect of temperature changes on chemical equilibrium systems

DSE 2014 Paper 1B Q13(a)(ii)

When a chemical equilibrium system is disturbed by a change in concentration, pressure or volume, the composition of the equilibrium mixture changes. However, as long as the temperature remains constant, the value of the equilibrium constant, K_c, does not change.

Reference Website
Visit the following website to explore the effect of a change in temperature on the equilibrium composition of a system:
http://www.chm.davidson.edu/vce/Equilibria/Temperature.html
(accessed April 2015)

Table 40.4 summarizes the effect of changing conditions on chemical equilibrium systems.

Table 40.4

The effect of changing conditions on chemical equilibrium systems

Change in	Composition of system	Value of K_c
concentration	changed	unchanged
pressure (by changing volume)	may change	unchanged
temperature	changed	changed

ASL 2007 Paper 2 Q4(a)(iii)

For a chemical equilibrium system involving a reaction with equal number of moles of gas on both sides of the equation, changing the pressure has no effect on the position of equilibrium, so the composition remains unchanged.

Notice that a temperature change nearly always changes the value of the equilibrium constant.

Effect of temperature change on an equilibrium system of hydrogen, iodine and hydrogen iodide

The reaction between hydrogen and iodine to give hydrogen iodide is exothermic.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \qquad \Delta H^{\ominus} = -10.4 \text{ kJ mol}^{-1}$$

A mixture of gaseous hydrogen and iodine is allowed to attain equilibrium at different temperatures. The equilibrium concentrations of the gases at these temperatures are listed in Table 40.5.

Table 40.5

The equilibrium concentrations of the gases at different temperatures		
Temperature (°C)	**$[H_2(g)]_{eqm}$ or $[I_2(g)]_{eqm}$ (mol dm^{-3})**	**$[HI(g)]_{eqm}$ (mol dm^{-3})**
25	0.033	0.934
230	0.068	0.864
430	0.107	0.786
490	0.114	0.773

We can see that as the temperature increases, the equilibrium concentration of the product decreases.

According to Le Chatelier's principle, the system responds by reducing the temperature upon a temperature increase. This brings about a net reaction that absorbs heat. As the backward reaction is endothermic, the system undergoes a net backward reaction, i.e. the position of equilibrium shifts to the left.

K_c and temperature changes

The equilibrium constant, K_c, for the reaction between hydrogen and iodide is as follows:

$$K_c = \frac{[HI(g)]^2}{[H_2(g)][I_2(g)]}$$

As the temperature increases, $[H_2(g)]$ and $[I_2(g)]$ increase while $[HI(g)]$ decreases. Thus, the equilibrium constant decreases. Table 40.6 lists the variation of K_c for the reaction with temperature. N5

N5 Increasing the temperature can shorten the time needed to attain equilibrium for all reversible reactions, but may NOT increase the yield.

Table 40.6

The variation of K_c for the reaction with temperature	
Temperature (°C)	**K_c (no units)**
25	801
230	161
430	54.0
490	46.0

In general, how the equilibrium constant, K_c, changes with a temperature change depends on the sign of $\Delta H^{\ominus}$ for the reaction.

- The value of K_c for an exothermic reaction (negative $\Delta H^{\ominus}$) decreases as the temperature increases.
- The value of K_c for an endothermic reaction (positive $\Delta H^{\ominus}$) increases as the temperature increases.

Effect of temperature change on an equilibrium system of dinitrogen tetroxide and nitrogen dioxide

40.2
Investigating the effect of pressure and temperature changes on equilibrium.

We can also use Le Chatelier's principle to predict the effect of temperature change on an equilibrium system. Take the endothermic decomposition of dinitrogen tetroxide (N_2O_4) as an example:

$$\underset{\text{pale yellow}}{N_2O_4(g)} \rightleftharpoons \underset{\text{dark brown}}{2NO_2(g)} \qquad \Delta H^{\ominus} = +57.2 \text{ kJ mol}^{-1}$$

Video
Investigating the effect of pressure and temperature changes on equilibrium

Three identical sealed tubes containing an equilibrium mixture of $N_2O_4(g)$ and $NO_2(g)$ are prepared at room temperature. They contain the same amount of $N_2O_4(g)$ and $NO_2(g)$ and they have the same brown colour.

One tube is now placed in ice-cold water. The second tube is kept at room temperature and the third tube is placed in hot water (Fig. 40.13).

Simulation
Effect of temperature changes on chemical equilibrium system

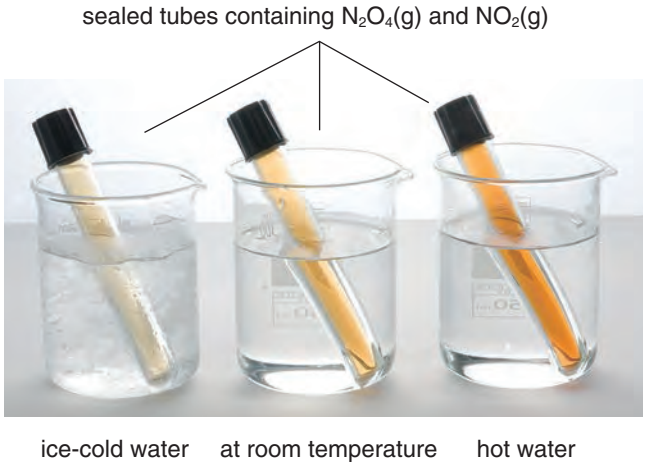


Fig. 40.13 The equilibrium mixture of $N_2O_4(g)$ and $NO_2(g)$ at different temperatures

Compared with the mixture kept at room temperature, the colour of the mixture in hot water is darker. This shows that more $NO_2(g)$ is present.

Upon a temperature increase, the system responds by reducing the temperature. This brings about a net reaction that absorbs heat. As the decomposition of $N_2O_4(g)$ is endothermic, the system undergoes a net forward reaction, i.e. the position of equilibrium shifts to the right. Thus, more $NO_2(g)$ is produced. So, the value of equilibrium constant, K_c, increases with increasing temperature.

Compared with the mixture kept at room temperature, the colour of the mixture in ice-cold water is much paler. This shows that more $N_2O_4(g)$ is present.

Upon a temperature decrease, the system responds by raising the temperature. This brings about a net reaction that releases heat. As the backward reaction is exothermic, the system undergoes a net backward reaction, i.e. the position of equilibrium shifts to the left. Thus, more $N_2O_4(g)$ is produced. So, the value of equilibrium constant, K_c, decreases with decreasing temperature. N6

Fig. 40.14 shows the effect of temperature change on the equilibrium constants for the two reversible reactions discussed.

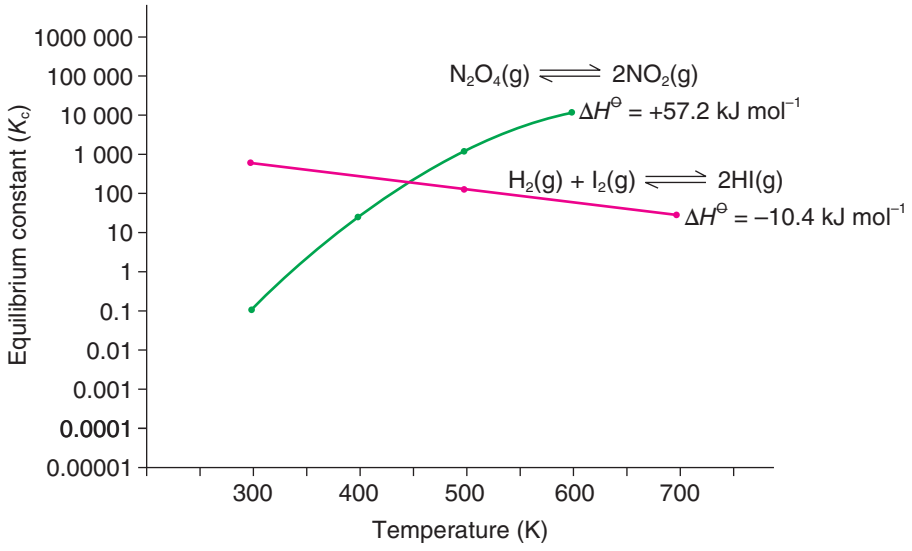


Fig. 40.14 The effect of temperature change on the equilibrium constants for the two reversible reactions discussed

N6 The effect of changes on the equilibrium system $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

Consider the following equilibrium system:

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$\Delta H > 0$

$K_c = [CO_2(g)]$

Effect of adding $CaCO_3(s)$

As the value of K_c depends only on temperature, adding $CaCO_3(s)$ to the system will not affect the concentration of $CO_2(g)$.

Effect of decreasing the temperature

The percentage decomposition of $CaCO_3(s)$ will decrease if the temperature of the equilibrium system is decreased.

The decomposition of $CaCO_3(s)$ is an endothermic process. A decrease in temperature will cause the equilibrium position to shift to the left.

N7 Refer to page T28.

Summary of effect of temperature changes on chemical equilibrium systems

Table 40.7 summarizes the effect of temperature changes on chemical equilibrium systems.

Table 40.7

A summary of effect of temperature changes on chemical equilibrium systems

Temperature change on equilibrium system	Rates of forward and backward reactions	Type of forward reaction	Direction of net reaction	Shift of position of equilibrium	Value of K_c
Temperature increase	rates of both reactions increase, but not to the same extent	exothermic	a net backward reaction occurs	to the left	decrease
		endothermic	a net forward reaction occurs	to the right	increase
Temperature decrease	rates of both reactions decrease, but not to the same extent	exothermic	a net forward reaction occurs	to the right	increase
		endothermic	a net backward reaction occurs	to the left	decrease

Worksheet N7

A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

The forward reaction is endothermic.

$K_c = \dfrac{[CO(g)][H_2O(g)]}{[H_2(g)][CO_2(g)]}$

From the data, we know that when the temperature is increased, the value of K_c increases. That means more CO(g) and $H_2O(g)$ are present in the equilibrium mixture. Thus, a net forward reaction occurs upon a temperature increase. According to Le Chatelier's principle, if the temperature of an equilibrium system is increased, the system responds by reducing the temperature. Thus, heat is absorbed when a net forward reaction occurs.

N8 Examination questions often list the values of K_c for a certain reaction at different temperatures and ask students to deduce whether the forward reaction is exothermic or endothermic.

Discussion

DSE 2014 Paper 1B Q13(b)

The table lists the equilibrium constant, K_c, for the reversible reaction:

$H_2(g) + CO_2(g) \rightleftharpoons CO(g) + H_2O(g)$

at three different temperatures.

Temperature (K)	K_c
500	7.76×10^{-3}
700	1.23×10^{-1}
900	6.03×10^{-1}

Based on the above information, deduce whether the forward reaction is exothermic or endothermic. Explain your answer using Le Chatelier's principle.

N8

Example 40.3

Q Consider the following reversible reaction:

$$Cu^{2+}(aq) + 4Cl^{-}(aq) \rightleftharpoons CuCl_4^{2-}(aq)$$
blue — yellow

A test tube containing a mixture of $Cu^{2+}(aq)$ ions and $CuCl_4^{2-}(aq)$ ions is allowed to attain equilibrium at room temperature. The test tube is then placed in ice-cold water. The blue colour of the mixture gradually becomes more intense.

a) State the effect of decreasing the temperature on the rate of the backward reaction.

b) Based on the given information, deduce whether the forward reaction is exothermic or endothermic.

A a) The rate of backward reaction decreases.

Upon a temperature decrease, the reactant particles have less kinetic energy and collide less often.

More importantly, the fraction of reactant particles with kinetic energy equal to or greater than the activation energy becomes smaller.

b) When the mixture is cooled, its blue colour becomes more intense, i.e. the concentration of $Cu^{2+}(aq)$ ions increases.

It can be deduced that upon a temperature decrease, the system undergoes a net backward reaction so as to raise the temperature.

Thus, the forward reaction is endothermic.

Practice 40.3

1 Nitrogen monoxide (NO) reacts with oxygen to give nitrogen dioxide (NO_2) in a closed vessel of fixed volume.

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g) \qquad \Delta H < 0$$

Suppose an equilibrium mixture of NO(g), $O_2(g)$ and $NO_2(g)$ is disturbed by a temperature increase. Predict the effect on:

a) the concentration of $NO_2(g)$; AL 2006 Paper 2 Q4(c)(ii)

b) the value of equilibrium constant, K_c; AL 2005 Paper 2 Q3(c)(ii)

c) the rate of formation of $NO_2(g)$.

Explain your answers.

a) Upon a temperature increase, the system responds by reducing the temperature. This brings about a net reaction that absorbs heat.
As the backward reaction is endothermic, the system undergoes a net backward reaction, i.e. the position of equilibrium shifts to the left, consuming some $NO_2(g)$. Thus, the concentration of $NO_2(g)$ decreases.
b) The value of equilibrium constant, K_c, decreases as the position of equilibrium shifts to the left.
c) When the temperature is increased, both the rates of the forward reaction and the backward reaction increase.
Thus, the rate of formation of $NO_2(g)$ increases.

Continued on next page

2 Consider the following reversible reaction:

$X(g) + Y(g) \rightleftharpoons Z(g) \qquad \Delta H < 0$

A mixture of X, Y and Z was allowed to attain equilibrium in a gas syringe.

Give the effects of each change, if any, on items listed in the table:

a) increasing the pressure; and

b) decreasing the temperature.

Effect of change on	(a) Increasing the pressure	(b) Decreasing the temperature
the rate of the forward reaction	increases	decreases
the rate of the backward reaction	increases	decreases
the position of equilibrium	to the right	to the right
value of K_c	no change	increases

40.7 Applying principles of equilibria and reaction rates to the Haber process

The principles of chemical equilibria and reaction rates play an important part in the choice of operating conditions for industrial processes. Manufacturers always want to:

- maximize the percentage conversion of reactants to products (increase the equilibrium yield);
- make this amount of product as quickly as possible (high reaction rate); and
- keep the costs as low as possible.

At the beginning of the twentieth century, European farmers imported the fertilizer sodium nitrate from Chile.



Fig. 40.15 Fritz Haber

In 1909, a German chemist Fritz Haber (Fig. 40.15) succeeded in synthesizing ammonia from nitrogen and hydrogen. The reaction was important because it allowed the manufacture of ammonia from nitrogen in the air. The ammonia could be used for the manufacture of ammonium salts for use as fertilizers (Fig. 40.16).

Now almost all ammonia is made by the **Haber process**, in which the reaction below is the key step.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92.2 \text{ kJ mol}^{-1}$$



Fig. 40.16 Farmers spraying fertilizers

What conditions will increase the yield of ammonia in the Haber process?

We can increase the percentage of ammonia in the equilibrium mixture by using:

- a low temperature; and
- a high pressure.

As the formation of ammonia is exothermic, decreasing the temperature will cause the position of equilibrium to shift to the right and thus increase the yield of ammonia (Fig. 40.17).

As there are 4 moles of gas on the left-hand side of the equation and 2 moles of gas on the right-hand side, increasing the pressure will cause the position of equilibrium to shift to the right and thus increase the yield of ammonia (Fig. 40.18).

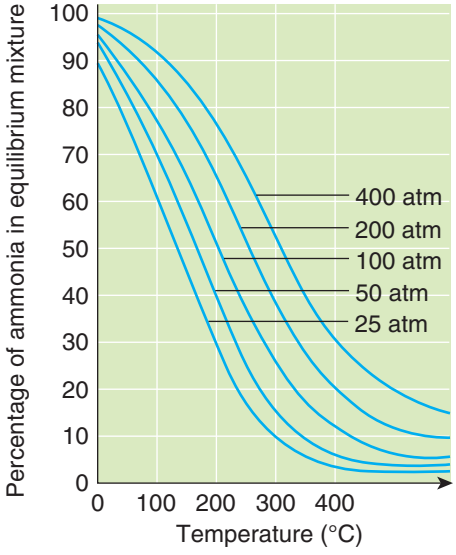


Fig. 40.17 The effect of temperature on the yield of ammonia

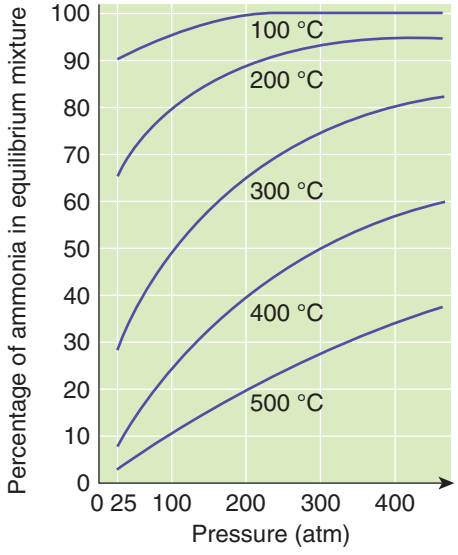


Fig. 40.18 The effect of pressure on the yield of ammonia

Haber process 哈柏法

However, there are drawbacks in using these theoretical conditions.

- Although a low temperature produces a high yield, the reaction will take place at a very low rate.
- A high pressure increases the concentrations of the gases and thus increases the rate. So, a high pressure produces both a high yield and rate. However, using a high pressure drastically increases the cost of building and running the plant. Large quantities of energy are required to compress the gases. There are also safety implications.

The pipes that carry the gaseous mixtures need to withstand a high pressure.

Any failure in the systems could allow chemicals to leak into the environment, endangering those working on site.

The operating conditions used in modern ammonia plants

In practice, a compromise is made between yield and rate.

- Temperature — This must be high enough to allow the reaction to proceed at a realistic rate while still producing an acceptable yield. A temperature between 400–450 °C is typically used.
- Pressure — A high pressure must still be used, but it must not be so high that the work force is put in danger or the environment threatened. A pressure of 200 atmospheres is typically used.
- Catalyst — An iron catalyst is added to speed up the rate of reaction, allowing the equilibrium to be attained faster and a lower temperature to be used. Less energy is used to generate heat, thus reducing the cost.

We will discuss how catalysts work in details in Topic 13 Industrial Chemistry.

DSE 2013 Paper 1A Q28

The catalyst increases the rates of both the forward reaction and the backward reaction to the same extent. It has no effect on the position of equilibrium and the yield. N9

Fig. 40.19 shows a diagram which outlines the Haber Process. Using the operating conditions outlined, the percentage yield of ammonia is about 20–30% in a single run. The ammonia produced is liquefied and removed. This shifts the position of equilibrium to the right.

We will further discuss the Haber process in Topic 13 Industrial Chemistry.

N9 Examination questions often ask about the effect of a catalyst on the rates of reactions and yeild.

Unreacted nitrogen and hydrogen are passed through the reactor again. Eventually, nearly all the nitrogen and hydrogen will be converted to ammonia.

The hydrogen used in the Haber process can be obtained from natural gas.

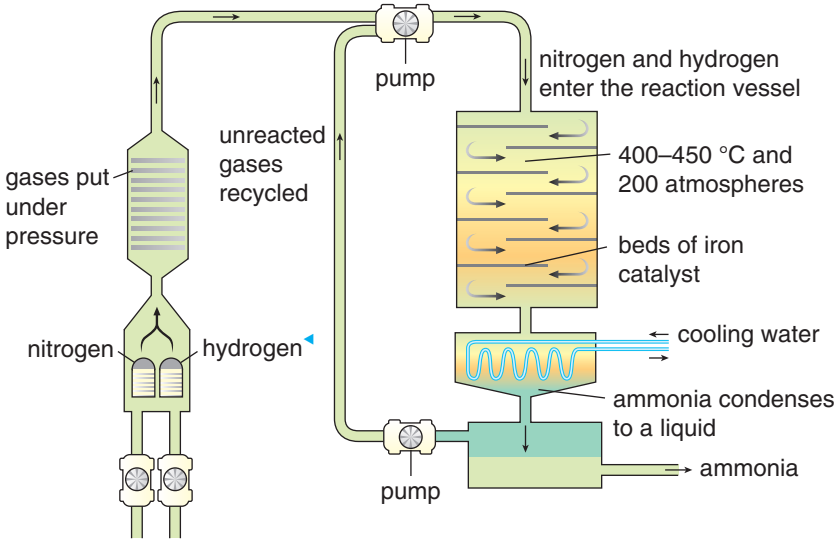


Fig. 40.19 A diagram for the Haber process

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Haber's discovery, which earned him a Nobel prize, was developed into a full-scale industrial process in 1914 (still used today). It is often claimed that the Haber process actually saved the world from starvation by making ammonia for fertilizer manufacture. On the other hand, without the discovery of this process, Germany would not have ammonia to make fertilizers (to increase the crop yield to feed its population) and explosives. Germany might have been unable to fight for so long during World War I.

Fritz Haber also managed to use chlorine and *mustard gas* as chemical weapons.

Discuss whether you think that Haber's invention contributed to the well-being of the world.

Discussion

Students thinking that Haber's inventions contributed to the well-being of the world may suggest:

- the use of fertilizers increases crop yield and reduces the chance of starvation;
- explosives have positive uses, e.g. construction, tunnelling.

Students thinking that Haber's inventions did not contribute to the well-being of the world may suggest:

- excess use of fertilizers may lead to environmental problems, e.g. nitrate contamination of water, eutrophication;
- explosives and chemical weapons cause deaths in warfare.

mustard gas 芥子氣

Example 40.4

Q Synthesis gas is a mixture of carbon monoxide and hydrogen. Methanol can be manufactured from synthesis gas in a reversible reaction as shown by the chemical equation below.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

A mixture of carbon monoxide and hydrogen is allowed to attain equilibrium in a closed container. Predict and explain how the time needed to attain equilibrium is affected by each of the following actions:

a) adding a catalyst;

b) increasing the temperature; and

c) increasing the volume of the container.

A a) The time needed is shorter.

A catalyst increases the rates of both the forward and backward reactions.

b) The time needed is shorter.

A higher temperature increases the rates of both the forward and backward reactions.

c) The time needed is longer.

In a larger container, the concentrations of the gases become smaller. Thus, the rates of both the forward and backward reactions decrease.

Practice 40.4

The hydrogen required for the Haber process is produced by reacting natural gas (methane) with steam at 850 °C and 30 atmospheres.

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$$
$$\Delta H = +210 \text{ kJ mol}^{-1}$$

a) Use Le Chatelier's principle to explain why such a high temperature is used.

b) What is the effect of using an increased pressure on the position of equilibrium?

c) Why is a higher pressure of 30 atmospheres used in the manufacturing process?

(*Hint*: Think about the rate of the reaction.)

Practice 40.4

1 a) According to Le Chatelier's principle, the system responds by reducing the temperature upon a temperature increase. This brings about a net reaction that absorbs heat.
As the production of hydrogen is endothermic, the system undergoes a net forward reaction, i.e. the position of equilibrium shifts to the right.
b) The system responds by reducing the pressure.
An increased pressure brings about a net reaction that decreases the number of moles of gas, i.e. the position of equilibrium shifts to the left.
c) The higher the pressure, the faster the reaction as there are more collisions in a unit volume per unit time.

40.8 Linking equilibria together

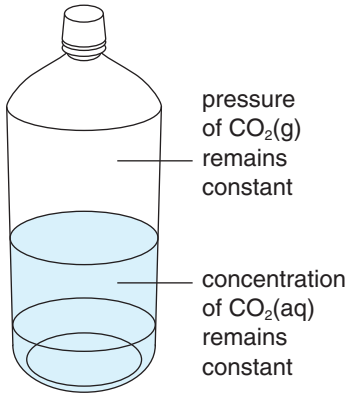


Fig. 40.20 A sealed bottle of soda water

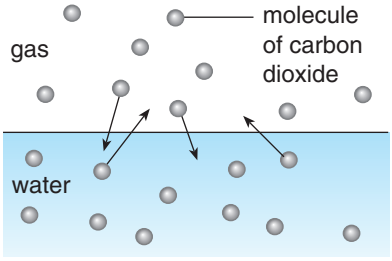


Fig. 40.21 In the system in the sealed bottle of soda water, the carbon dioxide molecules are entering and leaving the water

So far we have discussed chemical equilibrium systems involving different substances as reactants and products. However, equilibrium can also exist between two states of the same substance. Such an equilibrium is called a physical equilibrium because the changes that occur are physical processes.

Consider the sealed bottle of soda water shown in Fig. 40.20. In this system, carbon dioxide is present in two states: as a gas above the liquid, $CO_2(g)$, and as an aqueous solution in water, $CO_2(aq)$.

Suppose the bottle has been standing at a steady temperature for some time. We will find that the pressure of carbon dioxide gas above the water remains constant. The concentration of carbon dioxide dissolved in the water remains constant too. The system is at equilibrium. This is an example of physical equilibrium.

In the system, the carbon dioxide molecules are entering and leaving the water (Fig. 40.21). When the system reaches equilibrium, the molecules enter and leave at the same rate. This can be represented by the following equation:

$$CO_2(g) \rightleftharpoons CO_2(aq)$$

But when the carbon dioxide dissolves in the water, a small proportion of it actually reacts with the water instead of just dissolving. Hydrogencarbonate ions and hydrogen ions are formed.

$$CO_2(aq) + H_2O(l) \rightleftharpoons HCO_3^-(aq) + H^+(aq)$$

The product of the first equilibrium system ($CO_2(g) \rightleftharpoons CO_2(aq)$) is a reactant in the second, so that $CO_2(aq)$ is involved in both systems.

We can use Le Chatelier's principle to predict the effect of disturbing this equilibrium system by a change. For example, if more gaseous carbon dioxide is added, the position of the first equilibrium will shift to the right and more $CO_2(aq)$ will form. This increase in the concentration of $CO_2(aq)$ will then cause the position of the second equilibrium ($CO_2(aq) + H_2O(l) \rightleftharpoons HCO_3^-(aq) + H^+(aq)$) to shift to the right to produce more $HCO_3^-(aq)$ ions and $H^+(aq)$ ions.

Le Chatelier's principle helps in the development of theories to explain naturally occurring systems that involve reversible reactions, such as the dissolving of atmospheric carbon dioxide in the oceans.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Ice that burns

Methane hydrate (also known as fire ice) (Fig. 40.22) was discovered in the 1960s. It is often regarded as a combination of ice and methane. As the icy material melts, it releases the trapped methane. Hence, these hydrates represent a large energy reserve for humanity.



Fig. 40.22 Burning of methane hydrate

Methane hydrate consists of methane gas trapped inside cage-like crystal structures made up of water molecules (Fig. 40.23). Solid methane hydrate is in equilibrium with gaseous methane and water:

$$\text{methane hydrate(s)} \rightleftharpoons \text{methane(g)} + \text{water(l)} \qquad \Delta H > 0$$

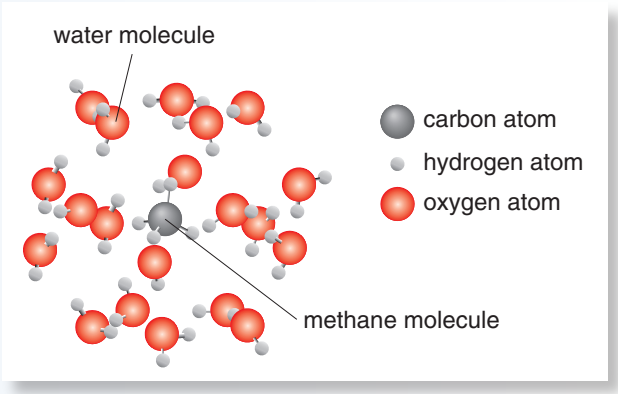


Fig. 40.23 Computer model of the structure of methane hydrate. The methane molecule is trapped within the space between water molecules

Methane hydrates are formed at the bottom of the sea where the water is cool and the pressure is high. Scientists estimate that the amount of methane trapped is between 2 and 10 times the known reserves of natural gas.

Questions

1 Explain the name 'ice that burns' for methane hydrate.

2 Predict and explain what would happen if the pressure on solid methane hydrate were reduced to 1 atmosphere.

3 Predict and explain what would happen to solid methane hydrate if it were heated.

4 Predict and explain what would happen if solid methane hydrate were brought to the surface of the sea.

1 Methane hydrate releases methane gas as it decomposes. Striking a match to some of the icy material releases water and methane. The methane catches fire and the heat from the flame can keep the process going. So, the icy material appears to melt and burn.

2 There is 1 mole of gas on the right-hand side of the equation but none on the left-hand side.
By Le Chatelier's principle, a pressure decrease would cause the position of equilibrium to shift to the side with a greater number of moles of gas, i.e. to the right in this case.
Thus, solid methane hydrate would decompose.

3 Upon a temperature increase, solid methane hydrate would respond by reducing the temperature, i.e. bringing about a net reaction that absorbs heat.
As the decomposition of solid methane hydrate is endothermic, a net forward reaction would occur, i.e. the position of equilibrium would shift to the right.
Thus, solid methane hydrate would decompose.

4 Raising solid methane hydrate from the sea bottom to the surface means a decrease in pressure and an increase in temperature.
As can be seen from the answers to questions 2 and 3, this would result in solid methane hydrate decomposing into methane gas and water.

Have you mastered?

Key terms

position of equilibrium 平衡位置 36 Le Chatelier's principle 勒沙得利爾原理 37

Haber process 哈柏法 59

Checklist

After studying this unit, you should be able to

- ☐ recognize that the value of K_c for an equilibrium system is a constant at constant temperature;
- ☐ deduce the effect of concentration change on the position of equilbirium of a system;
- ☐ deduce the effect of pressure change on the position of equilibrium of a system where a gas is involved;
- ☐ predict the effect of temperature change on the position of equilibrium of a system from the sign of ΔH for the forward reaction;
- ☐ derive the relation of temperature and the value of K_c from given data sets;
- ☐ recognize how manufacturers achieve a balance between the yield, the rate of reaction and the costs when choosing operating conditions for the Haber process.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Le Chatelier's principle states that if the condition of a system in equilibrium is changed, the position of equilibrium will shift so as to reduce that change.

2 If the concentration of one of the species in an equilibrium mixture is changed, the position of equilibrium will shift so as to reduce the change in concentration.

3 The following is a summary of effect of concentration changes on a chemical equilibrium system.

reactants $\rightleftharpoons$ products

Action on equilibrium system	Rates of forward and backward reactions	Direction of net reaction	Shift of position of equilibrium	Value of K_c
Increasing the concentration of a reactant	rate of forward reaction increases	a net forward reaction occurs	to the right	unchanged
Decreasing the concentration of a reactant	rate of forward reaction decreases	a net backward reaction occurs	to the left	
Increasing the concentration of a product	rate of backward reaction increases	a net backward reaction occurs	to the left	
Decreasing the concentration of a product	rate of backward reaction decreases	a net forward reaction occurs	to the right	

4 The following is a summary of effect of pressure changes on a chemical equilibrium system.

$aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$

Pressure (volume) change on equilibrium system		Rates of forward and backward reactions	Direction of net reaction	Shift of position of equilibrium	Value of K_c
Pressure increase (volume decrease)	a + b > c + d	rates of both reactions increase, but the forward reaction rate increases more	a net forward reaction occurs	to the right	unchanged
	a + b = c + d	same increase in rates of both reactions	no net reaction	no change	
	a + b < c + d	rates of both reactions increase, but the backward reaction rate increases more	a net backward reaction occurs	to the left	

Continued on next page

Pressure (volume) change on equilibrium system		Rates of forward and backward reactions	Direction of net reaction	Shift of position of equilibrium	Value of K_c
Pressure decrease (volume increase)	a + b > c + d	rates of both reactions decrease, but the forward reaction rate decreases more	a net backward reaction occurs	to the left	unchanged
	a + b = c + d	same decrease in rates of both reactions	no net reaction	no change	
	a + b < c + d	rates of both reactions decrease, but the backward reaction rate decreases more	a net forward reaction occurs	to the right	

5 The following is a summary of effect of temperature changes on a chemical equilibrium system.

Temperature change on equilibrium system	Rates of forward and backward reactions	Type of forward reaction	Direction of net reaction	Shift of position of equilibrium	Value of K_c
Temperature increase	rates of both reactions increase, but not to the same extent	exothermic	a net backward reaction occurs	to the left	decrease
		endothermic	a net forward reaction occurs	to the right	increase
Temperature decrease	rates of both reactions decrease, but not to the same extent	exothermic	a net forward reaction occurs	to the right	increase
		endothermic	a net backward reaction occurs	to the left	decrease

Have you mastered?

6 a) In the Haber process:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The percentage of ammonia in the equilibrium mixture can be increased by using

- a low temperature; and
- a high pressure.

b) 'Compromise' sets of operating conditions that manufacturers use for the Haber process:

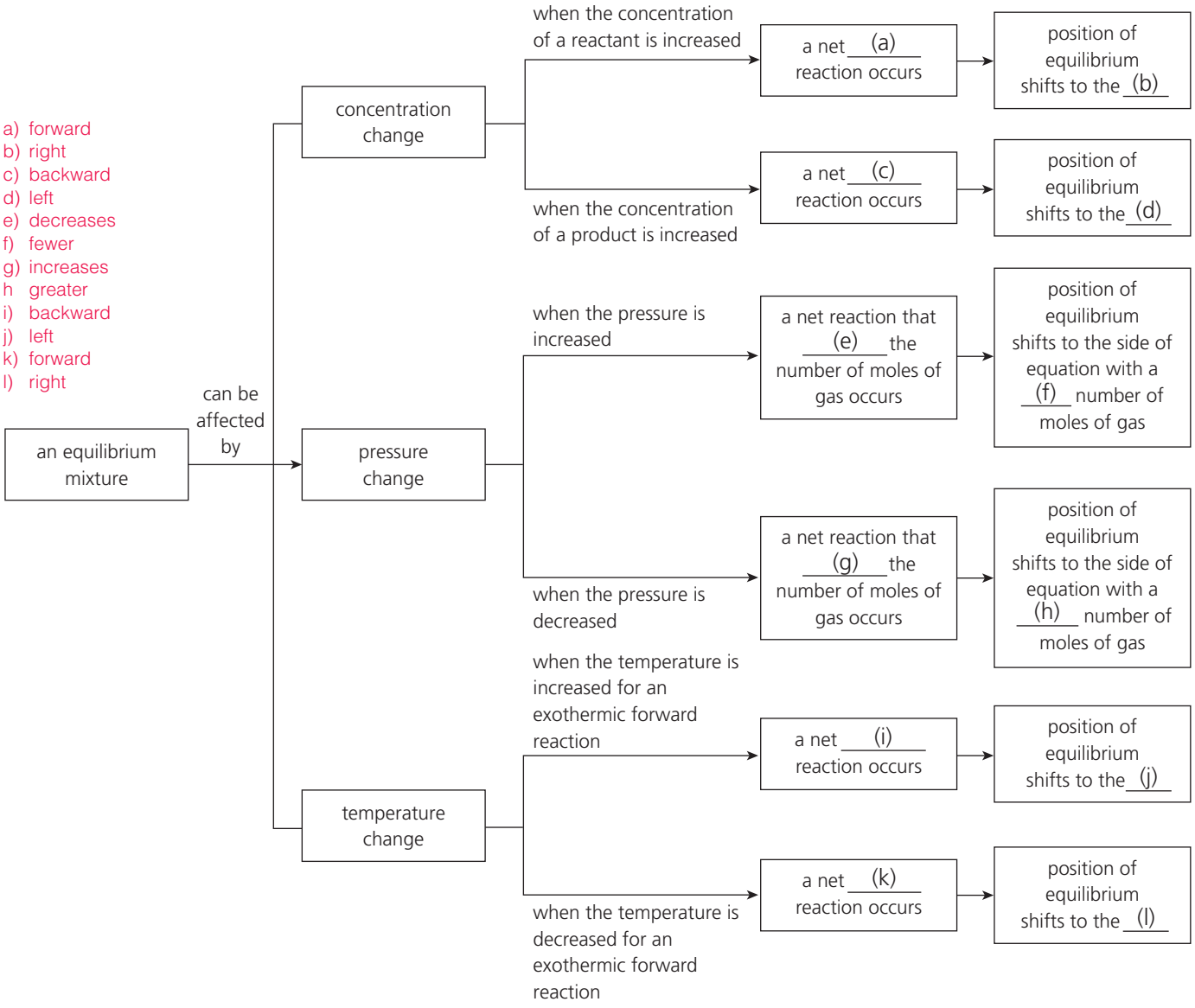
Temperature (°C)	Pressure (atm)	Catalyst
400–450	200	iron

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) forward
b) right
c) backward
d) left
e) decreases
f) fewer
g) increases
h greater
i) backward
j) left
k) forward
l) right

2 Consider the following reaction:

$H_2(g) + CO_2(g) \rightleftharpoons CO(g) + H_2O(g) \qquad \Delta H > 0$

How will the value of K_c be affected by

a) increasing the pressure (decreasing the volume)?
Not affected
b) increasing the temperature?
Increase
c) using a catalyst?
Not affected

3 Consider the following reaction:

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92 \text{ kJ mol}^{-1}$

What effect will each of the following changes have on the equilibrium concentration of $NH_3(g)$?

Change	Effect on $[NH_3(g)]_{eqm}$
a) Decreasing the temperature of the reaction vessel	increases
b) Increasing the concentration of $N_2(g)$	increases
c) Increasing the volume of the reaction vessel	decreases
d) Adding a catalyst	no effect

4 The first stage in the production of nitric acid is the oxidation of ammonia.

$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g) \qquad \Delta H = -910 \text{ kJ mol}^{-1}$

a) Complete the table to show what effect, if any, each of the changes in conditions will have on the position of equilibrium and on the rate at which equilibrium is attained.

Change in conditions	Effect on position of equilibrium	Effect on rate at which equilibrium is attained
Increasing the temperature	to the left	increases
Increasing the pressure	to the left	increases
Adding a catalyst	no effect	increases

b) Explain why, once the reaction has started, no further heating is required.
The reaction is exothermic.
(Edexcel GCE O Level, Paper1, Jan. 2010, 10(a), (c))

Part II Multiple choice questions

5 In the reaction between $Ag^+(aq)$ ions and $Fe^{2+}(aq)$ ions, what would be the effect of increasing the concentration of $Ag^+(aq)$ ions?

$Ag^+(aq) + Fe^{2+}(aq) \rightleftharpoons Fe^{3+}(aq) + Ag(s)$

A Rate of reaction increases; yield of $Fe^{3+}(aq)$ stays the same.
B Rate of reaction increases; yield of $Fe^{3+}(aq)$ decreases.
C Rate of reaction decreases; yield of $Fe^{3+}(aq)$ stays the same.
(D) Rate of reaction increases; yield of $Fe^{3+}(aq)$ increases.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2009, 13)

5 Increasing the concentration of $Ag^+(aq)$ ions
- increases the rate of reaction;
- shifts the position of equilibrium to the right, i.e. the yield of $Fe^{3+}(aq)$ increases.

6 A pressure increase shifts the position of equilibrium to the right, i.e. more pale yellow $N_2O_4(g)$ forms. The mixture of gases becomes paler brown.

Directions: Questions 6 and 7 refer to the following information.

Brown nitrogen dioxide, NO_2, exists in equilibrium with pale yellow dinitrogen tetroxide, N_2O_4.

$2NO_2(g) \rightleftharpoons N_2O_4(g) \qquad \Delta H = -57.2 \text{ kJ mol}^{-1}$
brown pale yellow

6 The pressure is increased. When equilibrium is restored, the appearance of the mixture of gases will be

A colourless.
B unchanged.
(C) paler brown.
D darker brown.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2012, 16(a))

Unit Exercise

7 The temperature is increased. When equilibrium is restored, the appearance of the mixture of gases will be

A colourless.
B unchanged.
C paler brown.
(D) darker brown.

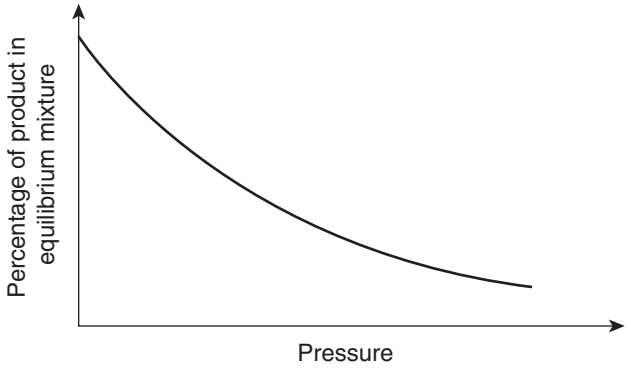
A temperature increase shifts the position of equilibrium to the left, i.e. more brown $NO_2(g)$ forms. The mixture of gases becomes darker brown.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2012, 16(b))

8 In which one of the following systems would the position of equilibrium NOT be affected by a volume change at a constant temperature?

A $2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$
B $C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$
C $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
(D) $CO(g) + H_2O(g) \rightleftharpoons H_2(g) + CO_2(g)$

9 The curve below shows the relationship between the percentage of product in the equilibrium mixture and pressure for a reaction.



Which of the following reactions will have the relationship shown?

A $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
(B) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
C $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$
D $CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$

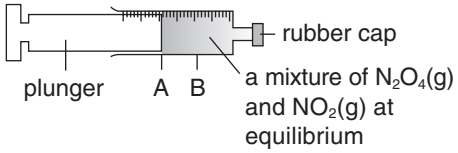
8 Consider $CO(g) + H_2O(g) \rightleftharpoons H_2(g) + CO_2(g)$
The number of moles of gas is the same on both sides of the equation, so changing the volume has NO effect on the position of equilibrium.

9 The graph shows that increasing the pressure would decrease the percentage of product in the equilibrium mixture.
Consider $N_2O4(g) \rightleftharpoons 2NO_2(g)$
A pressure increase brings about a net reaction that decreases the number of moles of gas, i.e. the position of equilibrium shifts to the reactant side. Thus, the percentage of product in the equilibrium mixture decreases.

11 A temperature increase shifts the position of equilibrium to the left. Thus, the yield of $SO_3(g)$ decreases.

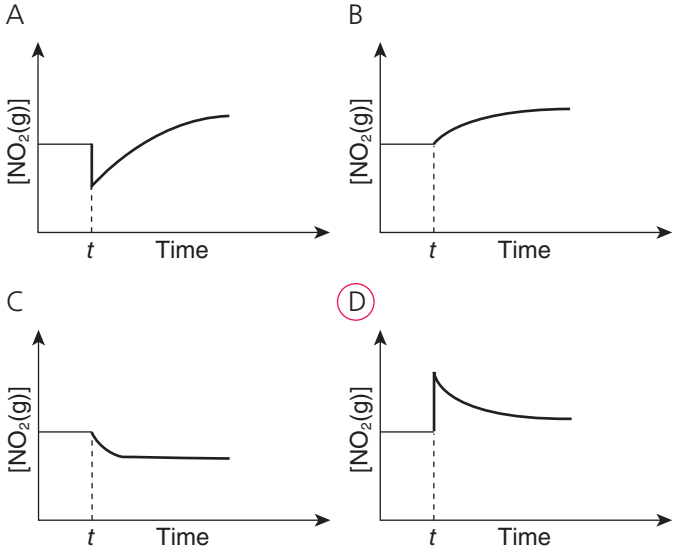
10 The diagram below shows a gas syringe containing a pale brown mixture of $N_2O_4(g)$ and $NO_2(g)$ at equilibrium.

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$
pale yellow dark brown



The plunger is quickly pushed from position A to position B at time *t*.

Which of the following graphs shows the variation of the concentration of $NO_2(g)$ in the mixture until a new equilibrium is attained?
Refer to page T28 for explanation.



11 A mixture of $SO_2(g)$, $O_2(g)$ and $SO_3(g)$ is allowed to attain equilibrium in a closed container.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\quad \Delta H < 0$

Which of the following combinations shows the effects of a temperature increase on the equilibrium system?

	Rate of forward reaction	Rate of backward reaction	Yield of $SO_3(g)$
A	increased	decreased	increased
(B)	increased	increased	decreased
C	decreased	increased	decreased
D	decreased	decreased	increased

12 1,2-dichloroethane decomposes in the presence of a catalyst.

$CH_2ClCH_2Cl(g) \rightleftharpoons CH_2{=}CHCl(g) + HCl(g)$

$\Delta H = +51\ kJ\ mol^{-1}$

Which of the following would result in an increase in the equilibrium yield of chloroethene?

(A) Increasing the temperature
B Increasing the pressure
C Increasing the surface area of the catalyst
D Changing the catalyst to a more efficient one

(Edexcel Advanced GCE, Unit 4, Jun. 2010, 16)

13 Ethanol can be manufactured by the reaction between ethane and water.

$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$

$\Delta H = -46\ kJ\ mol^{-1}$

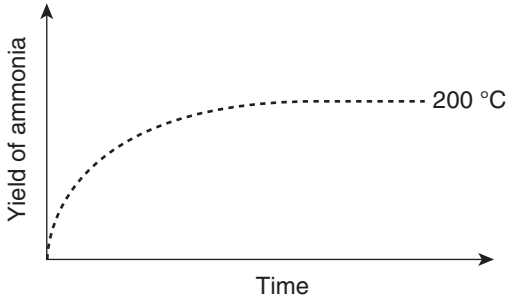
Which of the following sets of reaction conditions would produce the highest yield of ethanol?

A Low pressure and low temperature
(B) High pressure and low temperature
C Low pressure and high temperature
D High pressure and high temperature

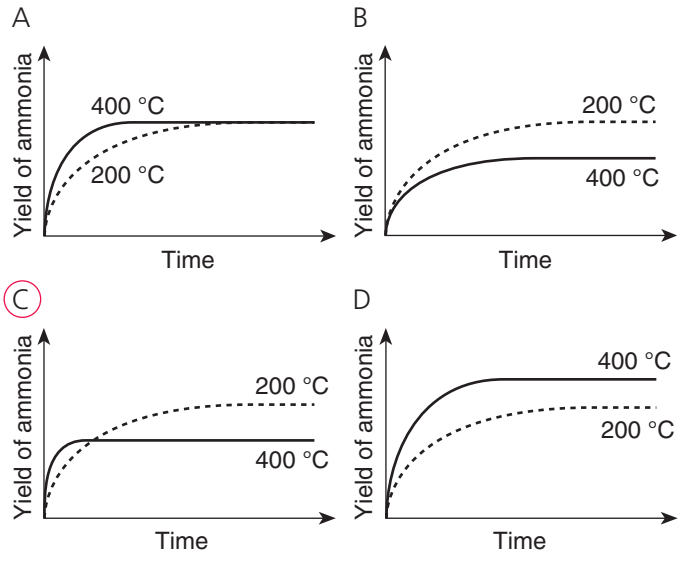
14 Ammonia is produced from hydrogen and nitrogen, according to the equation:

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H < 0$

The graph below shows the yield of ammonia at 200 °C and 100 atmospheres.



Which graph shows a correct comparison of the yield of ammonia at 400 °C and 100 atmospheres with the yield at 200 °C and 100 atmospheres?



15 A mixture of NO(g), $O_2(g)$ and $NO_2(g)$ is allowed to attain equilibrium in a closed container.

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

Which of the following actions will increase the concentration of $NO_2(g)$ in the mixture?

(1) Decreasing the volume of the container
(2) Removing some NO(g) from the container
(3) Adding an appropriate catalyst

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

12 Option A — Increasing the temperature shifts the position of equilibrium to the right, i.e. increases the equilibrium yield of chloroethene.
Options C and D — A catalyst has NO effect on the equilibrium yield of chloroethene.

13 A pressure increase shifts the position of equilibrium to the right, i.e. increases the yield of ethanol.
A temperature decrease shifts the position of equilibrium to the right, i.e. increases the yield of ethanol.

14 The equilibrium can be attained faster at 400 °C.
A temperature increase shifts the position of equilibrium to the left, i.e. decreases the yield of ammonia.

15 (1) Decreasing the volume of the container shifts the position of equilibrium to the right, i.e. the concentration of $NO_2(g)$ in the mixture will increase.
(2) Removing some NO(g) from the container shifts the position of equilibrium to the left, i.e. the concentration of $NO_2(g)$ in the mixture will decrease.
(3) Adding an appropriate catalyst has NO effect on the position of equilibrium, i.e. the concentration of $NO_2(g)$ in the mixture will remain the same.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

16 A mixture was prepared using 1.00 mole of propanoic acid, 2.00 moles of ethanol and 5.00 moles of water. At a given temperature, the mixture was left to reach equilibrium according to the following equation.

$$CH_3CH_2COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3CH_2COOCH_2CH_3(l) + H_2O(l)$$
$$\Delta H^{\ominus} = -22 \text{ kJ mol}^{-1}$$

The equilibrium mixture contained 0.540 mole of the ester ethyl propanoate.

Refer to page T29.

a) i) Calculate the amounts, in moles, of propanoic acid, of ethanol and of water in this equilibrium mixture.

0.460 mol; 1.46 mol; 5.54 mol

ii) Write an expression for the equilibrium constant, K_c, for the reaction.

iii) Calculate a value for K_c for the reaction at this temperatue. Explain why this K_c value has no units.

4.45

b) For this equilibrium, predict the effect of an increase in temperature on each of the following:

i) the amount, in moles, of ester at equilibrium;

ii) the time taken to attain equilibrium; and

iii) the value of K_c.

(AQA Advanced GCE, Unit 4, Jan. 2010, 1)

17 Ethene gas and bromine gas react according to the following equation:

$$C_2H_4(g) + Br_2(g) \rightleftharpoons C_2H_4Br_2(g)$$

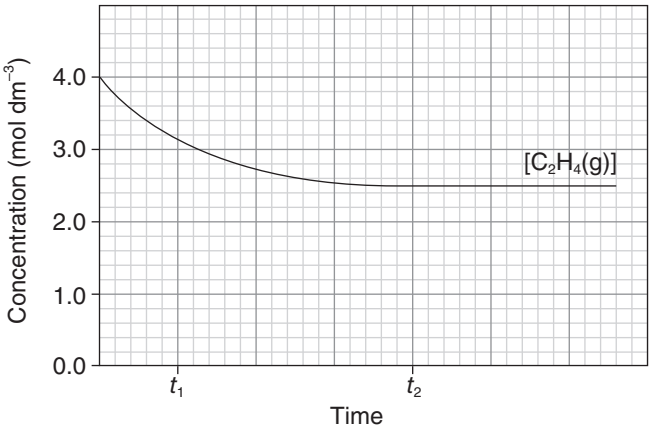
A chemist places 2.00 moles of $C_2H_4(g)$ and 1.25 moles of $Br_2(g)$ in a sealed 0.500 dm^3 container kept at a fixed temperature.

17 b) Number of moles of $C_2H_4(g)$ in the equilibrium mixture
= 2.50 mol dm^{-3} x 0.500 dm^3
= 1.25 mol
Number of moles of $C_2H_4(g)$ reacted = (2.00 – 1.25) mol
= 0.750 mol
According to the equation, 1 mole of C_2H_4 reacts with 1 mole of Br_2 to give 1 mole of $C_2H_4Br_2$.
i.e. number of moles of $Br_2(g)$ reacted = 0.750 mol
number of moles of $C_2H_4Br_2(g)$ formed = 0.750 mol

$$\text{Equilibrium concentration of } Br_2(g) = \frac{(1.25 - 0.750) \text{ mol}}{0.500 \text{ dm}^3} = 1.00 \text{ mol dm}^{-3}$$

$$\text{Equilibrium concentration of } C_2H_4Br_2(g) = \frac{0.750 \text{ mol}}{0.500 \text{ dm}^3} = 1.50 \text{ mol dm}^{-3}$$

The graph below shows the variation of the concentration of $C_2H_4(g)$ with time.



a) Compare the rate of the forward reaction and that of the backward reaction

i) at time t_1; At time t_1, the rate of forward reaction is greater than the rate of backward reaction.

ii) at time t_2. At time t_2, the rate of forward reaction is equal to the rate of backward reaction.

b) Determine the equilibrium concentrations of $Br_2(g)$ and $C_2H_4Br_2(g)$.

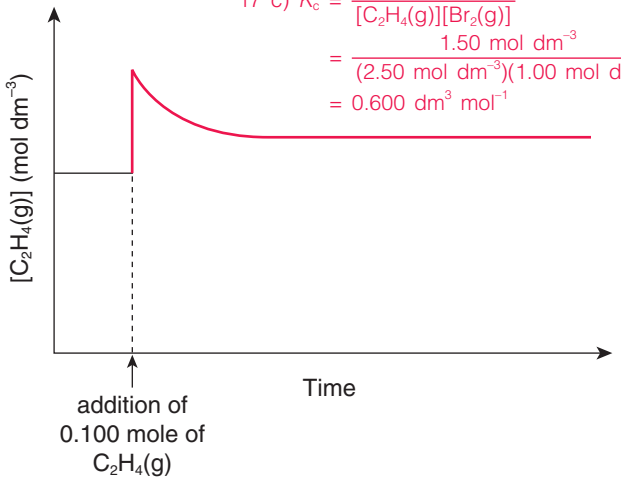
c) Calculate the equilibrium constant, K_c, for the reaction.

d) 0.100 mole of $C_2H_4(g)$ is added to the equilibrium mixture. Sketch, in the graph below, the variation of the concentration of $C_2H_4(g)$ with time until a new equilibrium is attained. (Assume that the temperature of the system remains the same throughout the whole process.)

17 c) $K_c = \dfrac{[C_2H_4Br_2(g)]}{[C_2H_4(g)][Br_2(g)]}$

$= \dfrac{1.50 \text{ mol dm}^{-3}}{(2.50 \text{ mol dm}^{-3})(1.00 \text{ mol dm}^{-3})}$

$= 0.600 \text{ dm}^3 \text{ mol}^{-1}$



Unit Exercise

18 Consider the reaction represented by the equation below:

$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons Fe(SCN)^{2+}(aq)$

In an experiment, 25.0 cm^3 of 0.010 M $Fe_2(SO_4)_3(aq)$ and 25.0 cm^3 of 0.010 M KSCN(aq) were mixed in a conical flask at room temperature, and equilibrium was attained.

a) The concentration of $Fe(SCN)^{2+}(aq)$ in the mixture was 0.0043 M when equilibrium was attained. Calculate the equilibrium constant, K_c, for the above reaction at room temperature.

b) It is known that $FePO_4(s)$ is insoluble in water. Suggest what would be the effect on the equilibrium position if $Na_3PO_4(s)$ is added to the equilibrium mixture.

(HKDSE, Paper 1B, 2012, 13)

19 Consider the reaction of $H_2(g)$ with $I_2(g)$ at elevated temperatues:

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

4.0 moles of $H_2(g)$ and 2.0 moles of $I_2(g)$ were introduced into an evacuated 5.0 dm^3 closed container kept at 713 K.

a) Given that the equilibrium constant, K_c, for the reaction is 50 at 713 K, calculate the concentrations of $H_2(g)$, $I_2(g)$ and HI(g), in mol dm^{-3}, respectively in the equilibrium mixture.

b) Deduce the effect on the number of moles of HI(g) in the equilibrium mixture

i) if the volume of the container is reduced to 2.5 dm^3;

ii) if the initial numbers of moles of $H_2(g)$ and $I_2(g)$ used are both 4.0.

(You may assume all other conditions to be the same in each case and are NOT required to carry out calculations.)

(HKASLE, Paper 2, 2008, 4(a))

20 Methanol can be manufactured from carbon monoxide and hydrogen. It is used as a fuel and for making chloroalkanes.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$ $\Delta H = -91$ kJ mol^{-1}

Refer to page T29.

a) The reaction represented by the above equation is an example of a dynamic equilibrium.

Explain what is meant by the term 'dynamic equilibrium'.

b) Describe and explain the effect of the following changes on the yield of methanol produced in the reaction:

i) carrying out the reaction at a higher temperature;

ii) increasing the total pressure of the reaction system.

c) Describe the effect that the use of a catalyst would have on the rate at which methanol is produced.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jun. 2011, 4(c))

21 Part of the Contact process for the manufacture of sulphuric acid involves the conversion of sulphur dioxide to sulphur trioxide, as shown by the equation:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -197$ kJ mol^{-1}

In a study of the process, a mixture of $SO_2(g)$, $O_2(g)$ and $SO_3(g)$ is allowed to reach dynamic equilibrium in a closed vessel. The reaction is catalyzed by V_2O_5.

Refer to page T30.

a) i) Explain the term 'dynamic equilibrium'.

iii) Give TWO other characteristics of a system in dynamic equilibrium.

b) What will happen to the position of equilibrium in each of the following cases? Explain in terms of Le Chatelier's principle.

i) Some $SO_3(g)$ is removed from the vessel.

ii) The pressure in the vessel is decreased.

iii) More V_2O_5 is added.

iv) The temperature of the vessel is increased.

22 Dinitrogen tetroxide is a pale yellow gas. It exists in equilibrium with nitrogen dioxide, a dark brown gas.

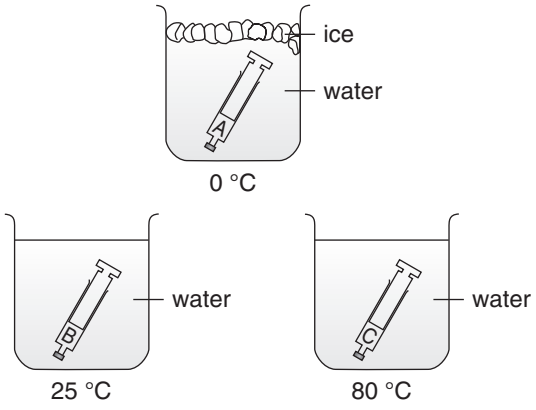
$$\underset{\text{pale yellow}}{N_2O_4(g)} \rightleftharpoons \underset{\text{dark brown}}{2NO_2(g)} \qquad \Delta H > 0$$

a) 0.400 mole of $N_2O_4(g)$ is heated in a closed reaction vessel of volume 16.0 dm^3. A state of equilibrium is attained at a constant temperature. The concentration of $NO_2(g)$ in the equilibrium mixture is found to be 0.0280 mol dm^{-3}.

i) What is the concentration of $N_2O_4(g)$ in the equilibrium mixture?

ii) Calculate the equilibrium constant, K_c, for the reaction.

b) There are three identical gas syringes, A, B and C, each containing an equilibrium mixture of $N_2O_4(g)$ and $NO_2(g)$ at 25 °C. Syringes A and C are then immersed in water at 0 °C and 80 °C respectively for 10 minutes.



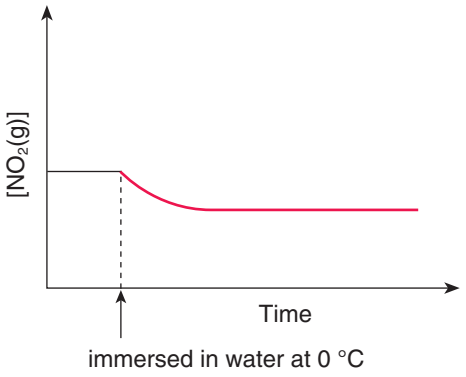
Compared with syringe B, state and explain the colour change of the mixture in each syringe.

Sketch on the given graph to show the expected variation in the concentration of $NO_2(g)$ in the mixture until the attainment of a new state of equilibrium.

i) Syringe A

(1) Colour change and explanation

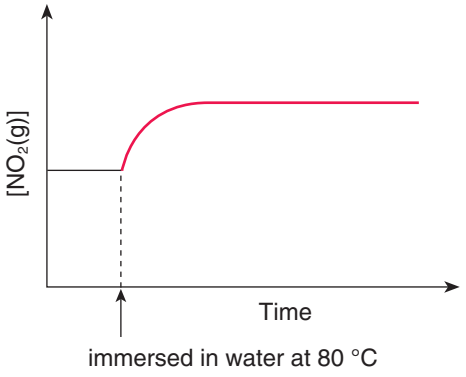
(2) Graph



ii) Syringe C

(1) Colour change and explanation

(2) Graph



22 a) i) Number of moles of $NO_2(g)$ produced
= 0.0280 mol dm^{-3} x 16.0 dm^3
= 0.448 mol

Number of moles of $N_2O_4(g)$ reacted = $\frac{0.448}{2}$ mol
= 0.224 mol

Number of moles of $N_2O_4(g)$ in the equilibrium mixture
= (0.400 – 0.224) mol
= 0.176 mol

Concentration of $N_2O_4(g)$ in the equilibrium mixture
= $\frac{0.176 \text{ mol}}{16.0 \text{ dm}^3}$
= 0.0110 mol dm^{-3}

ii) $K_c = \frac{[NO_2(g)]^2}{[N_2O_4(g)]}$
= $\frac{(0.0280 \text{ mol dm}^{-3})^2}{0.0110 \text{ mol dm}^{-3}}$
= 0.0713 mol dm^{-3}

b) i) (1) The colour of the mixture in syringe A becomes paler.
When the temperature is decreased, the system responds by raising the temperature.
As the backward reaction is exothermic, the system undergoes a net backward reaction.
More $N_2O_4(g)$ is produced.

ii) (1) The colour of the mixture in syringe C becomes darker.
When the temperature is increased, the system responds by reducing the temperature.
As the forward reaction is endothermic, the system will undergoes a net forward reaction.
More $NO_2(g)$ is produced.

23 Ethanol can be made industrially by reacting ethene with steam using a catalyst.

$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$

At various reaction conditions, the percentage conversion of ethene using excess steam at equilibrium is shown below.

Pressure (atm)	Temperature (°C)	Percentage conversion (%)
50	200	45
50	320	30
80	200	60
80	320	45

Refer to page T30.

a) State and explain the effect of increasing the pressure on the percentage conversion.

b) Use the data to deduce the sign of the enthalpy change for the forward reaction. Explain how you reach your conclusion.

c) The equation for the formation of ethanol shows that equal numbers of moles of ethene and steam are required. In industry, however, excess steam is used.

Suggest why excess steam is used.

(OCR Advanced Subsidiary GCE, Chemistry, 2813/01, Jun. 2009, 1(c))

24 Ozone, O_3, is formed when oxygen is exposed to ultraviolet (UV) radiation or to an electric discharge. Ozone is a blue gas whereas oxygen is colourless. When the two gases are mixed, an equilibrium is established as shown in the following equation.

$3O_2(g) \rightleftharpoons 2O_3(g) \quad \Delta H = +143 \text{ kJ mol}^{-1}$

Refer to page T31.

a) When the temperature of the pale blue equilibrium mixture is increased, the colour darkens. Explain this observation in terms of the changes to the equilibrium.

b) State and explain what you would see if the pressure of the system at equilibrium were increased.

c) A small amount of oxygen gas containing the isotope ^{18}O is added to the equilibrium mixture. After a few hours, ozone containing ^{18}O is detected. Given that the equilibrium position is NOT affected, explain this observation.

(Edexcel Advanced Subsidiary GCE, Unit 2, May 2012, 17(a))

25 In the manufacture of ammonia, hydrogen can be obtained from natural gas by reaction with steam.

$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$

$\Delta H = +206 \text{ kJ mol}^{-1}$

Refer to page T31.

a) A mixture of natural gas and steam was allowed to attain equilibrium. Explain how the time needed to attain equilibrium would be affected by

i) the use of a catalyst;

ii) a higher temperature;

iii) a lower pressure (by increasing the volume).

b) The mixture of natural gas and steam had attained equilibrium.

Explain how the composition of the equilibrium mixture would be affected by

i) the use of a catalyst;

ii) a higher temperature;

iii) a lower pressure (by increasing the volume).

26 An important reaction in the manufacture of nitric acid is the catalytic oxidation of ammonia.

$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$

$\Delta H = -910 \text{ kJ mol}^{-1}$

Refer to the next page.

a) Low pressures and low temperatures would give the maximum equilibrium yield of NO. Explain why.

b) The actual conditions used in the catalytic oxidation of ammonia include 900 °C and an increase in pressure.

Suggest why these conditions are a compromise.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2010, 6(a), (b))

27 Ammonia is manufactured by the Haber process:

$$N_2(g) + 3H_2(g) \overset{Fe(s)}{\rightleftharpoons} 2NH_3(g) \qquad \Delta H^{\ominus}_{298} = -92 \text{ kJ mol}^{-1}$$

In a simulation study of the process, mixtures of $N_2(g)$ and $H_2(g)$ were allowed to attain equilibrium under five sets of reaction conditions, and the mole percentage of $NH_3(g)$ in each equilibrium mixture was recorded. The table below lists the results obtained in the five trials:

Trial	Initial mole ratio of $N_2(g)$ to $H_2(g)$	Reaction conditions			Mole percentage of $NH_3(g)$ in equilibrium mixture
		Temperature (K)	Pressure (atm)	Catalyst	
1	1 : 3	473	10	Fe(s)	51
2	1 : 3	773	1 000	Fe(s)	58
3	1 : 3	473	1 000	Fe(s)	*a*
4	1 : 3	773	10	—	*b*
5	1 : 3	773	1 000	—	*c*

(No catalyst was used in trials 4 and 5; *a*, *b* and *c* represent the mole percentage of $NH_3(g)$ in the equilibrium mixture in trials 3, 4 and 5 respectively.)

a) In which TWO trials would the mole percentage of $NH_3(g)$ in the equilibriium mixture be the same? Explain your answer.

b) In which trial would the mole percentage of $NH_3(g)$ in the equilibrium mixture be the highest? Explain your answer.

c) The industrial operating conditions for the Haber process are as follows:

Mole ratio of $N_2(g)$ to $H_2(g)$	1:3
Temperature	673 K
Pressure	200 atm
Catalyst	Fe(s)

Explain why this set of conditions is used.

(HKASLE, Paper 2, 2011, 6)

26 a) A pressure decrease brings about a net reaction that increases the number of moles of gas. The position of equilibrium shifts to the side of the equation with a greater number of moles of gas, i.e. to the right.
A temperature decrease brings about a net reaction that releases heat. As the forward reaction is exothermic, the system undergoes a net forward reaction, i.e. the position of equilibrium shifts to the right.
b) The 900 °C temperature increases the rate of the reaction while still produces an acceptable yield.
The increased pressure increases the rate of the reaction while still produces an acceptable yield.

28 Methanol can be manufactured from carbon monoxide and hydrogen.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \qquad \Delta H < 0$

A mixture of CO(g), H_2(g) and CH_3OH(g) was allowed to attain equilibrium in a cylinder fitted with a movable piston. Give the effect of each change on items listed in the following table:

a) adding a catalyst;

b) decreasing the temperature;

c) increasing the pressure by decreasing the volume.

Effect of change on	a) Adding a catalyst	b) Decreasing the temperature	c) Increasing the pressure
the rate of the forward reaction	increases	decreases	increases
the rate of the backward reaction	increases	decreases	increases
the position of equilibrium	no effect	shifts to the right	shifts to the right
the equilibrium constant, K_c	no effect	increases	no effect

4 A pressure increase (i.e. a volume decrease) increases the rates of the forward and backward reactions.
A pressure increase (i.e. a volume decrease) shifts the position of equilibrium to the right, i.e. increases the yield of N_2F_4(g).

5 The observation indicates that the position of equilibrium shifts to the right upon a temperature decrease.
A temperature decrease brings about a net reaction that release heat. It can be deduced that the forward reaction is exothermic.
The concentration of deep red $Fe(SCN)^{2+}$(aq) ions and the value of K_c have increased.

Topic Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Multiple choice questions

1 Hydrogen and iodine, both with an initial concentration of 0.010 mol dm^{-3}, were allowed to react. At equilibrium, the concentration of hydrogen iodide was 0.0030 mol dm^{-3}.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

K_c is calculated using the values
Refer to page T32 for explanation.

	$[H_2(g)]_{eqm}$ (mol dm^{-3})	$[I_2(g)]_{eqm}$ (mol dm^{-3})	$[HI(g)]_{eqm}$ (mol dm^{-3})
A	0.0070	0.0070	0.0030
B	0.0040	0.0040	0.0030
C	0.0040	0.0040	0.0060
(D)	0.0085	0.0085	0.0030

(Edexcel Advanced GCE, Unit 4, Jan. 2012, 2)

2 A mixture of $N_2O_4(g)$ and $NO_2(g)$ is allowed to attain equilibrium in a gas syringe at room temperature:

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$
pale yellow — dark brown

The gas mixture in the syringe is compressed rapidly. Which of the following statements correctly describes the expected observation?

A The colour of the mixture becomes paler.
B The colour of the mixture becomes darker.
C The colour of the mixture becomes paler instantaneously and then darker.
(D) The colour of the mixture becomes darker instantaneously and then paler.

(HKDSE, Practice paper 1A, 2012, 26)

3 Which of these will NOT improve the overall yield of the Haber process?

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -92$ KJ mol^{-1}

3 Increasing the temperature shifts the position of equilibrium to the left, i.e. will NOT improve the overall yield of the Haber process.

A Increasing the pressure
B Liquefying then removing the ammonia from the reaction
(C) Increasing the temperature
D Recycling unreacted nitrogen and hydrogen

(Edexcel Advanced GCE, Unit 4, Jun. 2012, 4)

4 Consider the following equilibrium system:

$2NF_2(g) \rightleftharpoons N_2F_4(g)$

Which of the following combinations shows the effects on the equilibrium system when the volume of the reaction vessel is decreased?
Refer to the previous page for explanation.

	Rate of forward reaction	Rate of backward reaction	Yield of $N_2F_4(g)$
(A)	increased	increased	increased
B	increased	decreased	decreased
C	decreased	decreased	decreased
D	decreased	increased	increased

🌶🌶5 Consider the following equilibrium system:

$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons Fe(SCN)^{2+}(aq)$
yellow-brown — colourless — deep red

In an experiment, a student places the above equilibrium system into a cold water bath and notes that the intensity of the red colour increases.

It can be deduced that the forward reaction is
Refer to the previous page for explanation.

(A) exothermic and the value of K_c has increased.
B exothermic and the value of K_c has decreased.
C endothermic and the value of K_c has increased.
D endothermic and the value of K_c has decreased.

Directions: Questions 6 and 7 refer to the following reversible reaction:

$$X_2(g) + 3Y_2(g) \rightleftharpoons 2XY_3(g)$$

6 A mixture of $X_2(g)$ and $Y_2(g)$ was introduced into a 2.0 dm^3 closed vessel kept at a fixed temperature. When the system attained equilibrium, the vessel contained 0.4 mole of $X_2(g)$, 0.3 mole of $Y_2(g)$ and 0.4 mole of $XY_3(g)$.

Which of the following is the numerical value of K_c for the above reaction at this temperature?

A 3.3
B 6.7
C 14.8
(D) 59.3

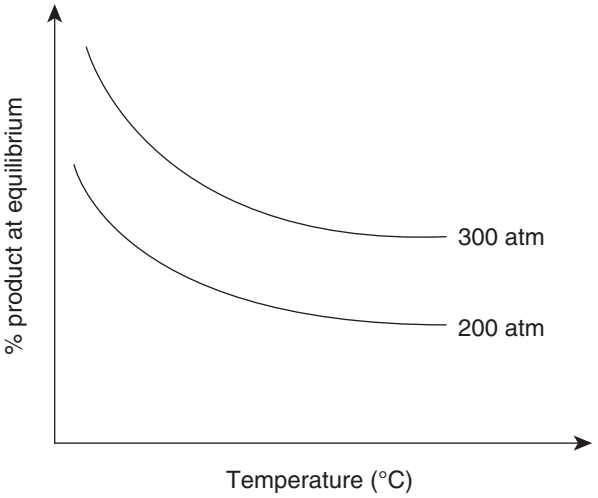
(HKDSE, Paper 1A, 2013, 27)

7 Which of the following combinations shows the effects of a catalyst on the rate of forward reaction, rate of backward reaction and the yield of $XY_3(g)$?

	Rate of forward reaction	Rate of backward reaction	Yield of $XY_3(g)$
(A)	increased	increased	unchanged
B	unchanged	unchanged	unchanged
C	increased	decreased	increased
D	decreased	increased	decreased

(HKDSE, Paper 1A, 2013, 28)

8 The graph below shows the yield of product in a gaseous equilibrium at different temperatures and pressures.



The forward reaction in the equilibrium system is

A exothermic, and the number of moles of gas is increasing.
B endothermic, and the number of moles of gas is increasing.
(C) exothermic, and the number of moles of gas is decreasing.
D endothermic, and the number of moles of gas is decreasing.

(Edexcel Advanced GCE, Unit 4, Jan. 2011, 6)

9 Which of the following is / are characteristic(s) of chemical equilibrium?

(1) When a catalyst is added to an equilibrium mixture, the equilibrium position changes.
(2) When equilibrium is attained, the rate of forward reaction and that of backward reaction are equal.
(3) Equilibrium can be attained from either direction of the reaction.

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKDSE, Practice paper 1A, 2012, 31)

8 From the graph, it can be deduced that the yield of product decreases with a temperature increase. A temperature increase shifts the position of equilibrium to the left. Thus, the forward reaction should be exothermic.
From the graph, it can be deduced that the yield of product increases with a pressure increase. A pressure increase shifts the position of equilibrium to the right. Thus, the number of moles of gas is decreasing.

Directions : *Each question (Questions 10–12) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :*

A	Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B	Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C	The 1st statement is false but the 2nd statement is true.
D	Both statements are false.

	1st statement	2nd statement	
10	A system with Q_c much greater than K_c can attain equilibrium rapidly. The difference between Q_c and K_c can be used to predict the direction of a reaction, NOT the rate at which equilibrium is attained.	The difference between Q_c and K_c can be used to predict the rate at which equilibrium is attained.	D
11	Increasing the pressure can increase the yield for all reversible chemical reactions. Increasing the pressure does NOT always cause the position of equilibrium of a system to shift to the right.	Increasing the pressure can cause the position of equilibrium of a system to shift to the right.	D
12	Increasing reaction temperature can increase the yield for all reversible chemical reactions. Increasing reaction temperature does NOT always increase the yield of a reversible chemical reaction.	Increasing reaction temperature can shorten the time needed to attain equilibrium for all reversible chemical reactions.	C

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

13 Ethanol is manufactured by the catalytic hydration of ethene:

$$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$$

The equilibrium constant for the reaction at 573 K is 0.115 $dm^3\ mol^{-1}$.

A mixture of 2.00 moles of $C_2H_4(g)$, 1.00 mole of $H_2O(g)$ and 0.100 mole of $C_2H_5OH(g)$ is introduced into an evacuated vessel of 1.20 dm^3 kept at 573 K.
Refer to page T32.

a) For this system under the initial conditions, calculate its reaction quotient.
0.0600 $dm^3\ mol^{-1}$

b) Predict and explain, under the initial conditions, whether the forward reaction rate or the backward reaction rate is greater.

c) Calculate the concentration of $C_2H_5OH(g)$ when the system attains equilibrium.
0.143 $mol\ dm^{-3}$

14 Account for the following observations and give the relevant chemical equation(s):

Words written on a paper using KSCN(aq) are invisible. When the paper is sprayed with $Fe^{3+}(aq)$, the words appear blood-red. If the words are written with alkaline KSCN(aq), they will turn orange-brown when sprayed with $Fe^{3+}(aq)$.

(HKASLE, Paper 1, 2012, 1(b))

15 At 250 °C, the equilibrium constant, K_c, for the following reaction is 25 mol^{-1} dm^3.

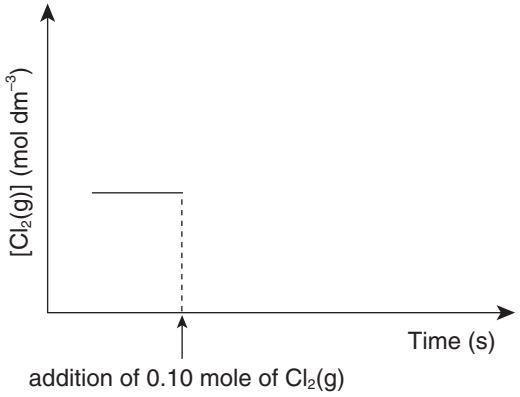
$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

A 10.0 dm^3 sealed container, which is maintained at 250 °C, initially contains 0.50 mole of $PCl_3(g)$, 0.20 mole of $Cl_2(g)$ and 0.40 mole of $PCl_5(g)$.

a) For this system under the initial conditions, calculate its reaction quotient. Predict and explain, under the initial conditions, whether the forward reaction rate or the backward reaction rate would be greater.

b) Calculate the concentration of $Cl_2(g)$ when the system attains equilibrium at 250 °C.

c) 0.10 mole of $Cl_2(g)$ is added to the equilibrium mixture in (b). Sketch, in the graph below, the variation of the concentration of $Cl_2(g)$ with time until a new equilibrium is attained. (Assume that the temperature of the system remains at 250 °C throughout the whole process.)



(HKDSE, Paper 1B, 2013, 12)

16 Under suitable coniditons, the equilibrium represented below was established.

$2CH_4(g) \rightleftharpoons 3H_2(g) + C_2H_2(g)$ $\quad \Delta H^\ominus = +377$ kJ mol^{-1}

Refer to page T33.

a) Write an expression for the equilibrium constant, K_c, for this reaction.

b) At a given temperature and pressure, the equilibrium mixture contained 0.440 mole of methane, 0.280 mole of hydrogen and 0.120 mole of ethyne (C_2H_2) in a container of volume 0.250 dm^3.

Calculate the value of K_c under these conditions and deduce its units. 0.218 mol^2 dm^{-6}

c) State the effect of an increase in temperature on the position of equilibrium and on the value of K_c for this reaction.

d) State the effect of an increase in pressure on the position of equilibrium and on the value of K_c for this reaction.

(AQA Advanced GCE, Unit 4, Jan. 2008, 3(a)–(d))

17 At high temperatures and in the presence of a catalyst, sulphur trioxide decomposes according to the following equation:

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ $\quad \Delta H^\ominus = +196$ kJ mol^{-1}

Refer to page T33.

a) In an experiment, 8.00 mole of sulphur trioxide were placed in a container of volume 12.0 dm^3 and allowed to come to equilibrium.

At temperature T_1, there were 1.40 moles of oxygen in the equilibrium mixture.

i) Calculate the amount, in moles, of sulphur trioxide and of sulphur dioxide in the equilibrium mixture.
5.20 mol; 2.80 mol

ii) Write an expression for the equilibrium constant, K_c, for the reaction.

iii) Deduce the units of K_c for the reaction.

iv) Calculate a value of K_c for the reaction at temperature T_1.
0.0338 mol dm^{-3}

b) The experiment was repeated at the same temperature using the same amount of sulphur trioxide but in a larger container.

State the effect, if any, of this change on:

i) the amount, in moles, of oxygen in the new equilibrium mixture;

ii) the value of K_c.

c) The experiment was repeated in the original container but at temperature T_2.

The new value of K_c was smaller than the value at temperature T_1.

State which is the higher temperature, T_1 or T_2. Explain your answer.

(AQA Advanced GCE, Unit 4, Jan. 2012, 2)

18 Nitric acid, HNO_3 is manufactured in large quantities. The main use of nitric acid is in the manufacture of fertilizers.

In its industrial preparation, nitric acid is produced in three main stages.

Stage 1

Ammonia is heated with oxygen in the air to form nitrogen monoxide, NO, and steam.

Stage 2

The hot nitrogen monoxide gas is then mixed with air and cooled under pressure. Nitrogen dioxide, NO_2, forms in the reversible reaction:

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g) \quad \Delta H = -115 \text{ kJ mol}^{-1}$

Stage 3

The nitrogen dioxide is reacted with water in a series of reactions to form nitric acid, HNO_3. The first of these reactions forms a mixture of nitric acid, HNO_3, and nitrous acid, HNO_2.

Refer to page T34.

a) In *Stage 2*, explain why the equilibrium mixture is both cooled and put under pressure.

b) Construct an equation for

- the reaction that takes place in *Stage 1*;
- the first reaction that takes place in *Stage 3*.

c) An industrial chemist carries out some research into the equilibrium system used in *Stage 2* of the manufacture of nitric acid.

- The chemist mixes together 0.800 mole of NO(g) and 0.700 mole of $O_2(g)$ in a container with a volume of 2.00 dm^3.
- The chemist heats the mixture and allows it to stand at constant temperature to attain equilibrium.

  The container is kept under pressure so that the total volume is maintained at 2.00 dm^3.
- At equilibrium, 75.0% of the NO(g) has reacted.

  i) Write an expression for K_c for the reaction.

  ii) Calculate the equilibrium constant, K_c, including units, for the reaction. 45.0 dm^3 mol^{-1}

(OCR Advanced GCE, Chem. A, F325, Jun. 2011, 2)

19 Nitrogen monoxide is an atmospheric pollutant, formed inside car engines by the reaction between nitrogen and oxygen.

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g) \quad \Delta H = +66 \text{ kJ mol}^{-1}$

This reaction is endothermic.

A research chemist investigates the reaction between nitrogen and oxygen.

She mixes nitrogen and oxygen gases in a sealed container. She then heats the container at a constant temperature for one day until the gases reach a dynamic equilibrium.

a) Explain, in terms of the rate of the forward reaction and the rate of the backward reaction, how the mixture of $N_2(g)$ and $O_2(g)$ reaches a dynamic equilibrium containing $N_2(g)$, $O_2(g)$ and NO(g).

b) The research chemist repeats the experiment at the same temperature using the same initial amounts of $N_2(g)$ and $O_2(g)$. This time she carries out the experiment at a much higher pressure.

Suggest why

- much less time is needed to reach dynamic equilibrium; and
- the composition of the equilibrium mixture is the same as in the first experiment.

c) The reaction between nitrogen and oxygen in a car engine does NOT reach a dynamic equilibrium. Suggest why not.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2011, 3(c))

19 a) Rate of forward reaction decreases while rate of backward reaction increases, until the rate of forward reaction equals the rate of backward reaction.

b) Increasing the pressure means increasing the concentrations of the reactants.
Thus, the rates of both the forward and backward reactions increase.
A pressure change does not change the composition of the equilibrium mixture because the number of moles of gas is the same on both sides of the equation.

c) It is not a closed system.

20 In an experiment, excess aqueous ammonia is added to an aqueous solution of copper(II) sulphate. The following equilibrium is established and the resulting solution is deep blue in colour.

$Cu^{2+}(aq) + 4NH_3(aq) \rightleftharpoons Cu(NH_3)_4^{2+}(aq)$

a) Write an expression of K_c for this reaction.

b) If the above equilibrium mixture contains 0.0020 mol dm^{-3} of $Cu^{2+}(aq)$ ions, 0.0014 mol dm^{-3} of $NH_3(aq)$ and 0.0800 mol dm^{-3} of $Cu(NH_3)_4^{2+}(aq)$ ions, calculate K_c under the conditions of the experiment.

c) When $H_2SO_4(aq)$ is added slowly to the equilibrium mixture until in excess, a blue precipitate is formed and the precipitate subsequently dissolves in the excess acid forming a blue solution. Account for these observations with the help of relevant chemical equation(s).

(HKDSE, Practice paper 1B, 2012, 13)

21 Carbon monoxide and hydrogen are used in the manufacture of methanol. An equilibrium system is established according to the following equation:

$$CO(g) + 2H_2(g) \overset{\text{Cu catalyst}}{\rightleftharpoons} CH_3OH(g) \quad \Delta H = -91 \text{ kJ mol}^{-1}$$

Refer to page T35.

a) Give TWO features of a reaction system at equilibrium.

b) Explain why an increase in temperature causes a decrease in the equilibrium yield of methanol.

c) i) State what is meant by the term 'catalyst'.

ii) State the effect, if any, of the copper catalyst on the position of this equilibrium at a fixed temperature.

(AQA Advanced Subsidiary GCE, Unit 2, Jun. 2009, 2(a)–(c))

22 Ammonia is manufactured by the Haber process in which the following equilibrium system is established.

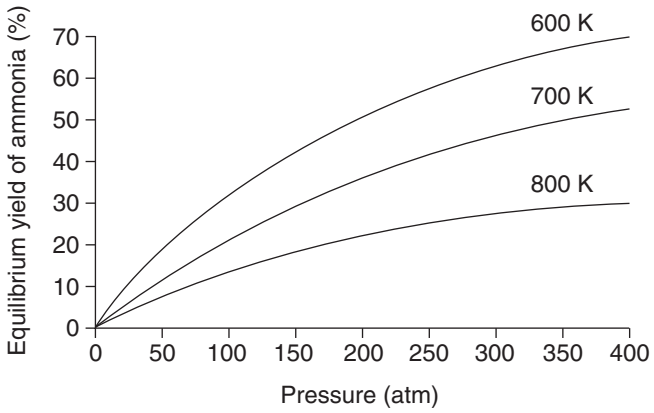
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Refer to page T35.

a) Give TWO features of a reaction system at equilibrium.

b) Explain why a catalyst has NO effect on the equilibrium position of an equilibrium system.

c) The diagram shows how the equilibrium yield of ammonia varies with changes in pressure and temperature.



i) Use the diagram to state the effect of an increase in pressure at constant temperature on the yield of ammonia. Use Le Chatelier's principle to explain this effect.

ii) Use the diagram to state the effect of an increase in temperature at constant pressure on the yield of ammonia. Use Le Chatelier's principle to explain this effect.

d) At equilibrium, with a pressure of 350 atmospheres and a temperature of 600 K, the percentage yield of ammonia is 65%.

i) State why industry uses a temperature higher than 600 K.

ii) State why industry uses a pressure lower than 350 atmospheres. (Do NOT include references to safety.)

(AQA Advanced Subsidiary GCE, Unit 2, Jan. 2012, 2)

23 Hydrogen has many industrial uses including making margarine and ammonia.

Hydrogen can be made by the reaction between methane and steam.

$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$

$\Delta H = +210 \text{ kJ mol}^{-1}$

Refer to page T36.

a) The pressure of the equilibrium mixture is increased.

Explain what happens to the position of the equilibrium.

b) The temperature of the equilibrium mixture is increased.

Explain what happens to the position of the equilibrium.

c) The reaction is actually carried out in the presence of a nickel catalyst at a pressure of 30 atmospheres.

Suggest why the manufacturer uses a pressure of 30 atmospheres.

(OCR Advanced Subsidiary GCE, Chem. A, F322, May 2012, 3(a)–(c)(i))

24 A chemist carries out an investigation on the equilibrium system shown below.

$$2CO(g) + 2NO(g) \rightleftharpoons 2CO_2(g) + N_2(g)$$
$$\Delta H = -788 \text{ kJ mol}^{-1}$$

The chemist mixes 0.460 mole of CO with 0.450 mole of NO. The mixture is left to attain equilibrium at constant temperature.

The student analyzes the equilibrium mixture and finds that 0.250 mole of NO remains. The total volume of the equilibrium mixture is 1.00 dm^3.

Refer to page T36.

a) i) Write the K_c expression for the reaction.

ii) What are the units of this equilibrium constant?

iii) Determine the value of K_c for the reaction.
0.947 dm^3 mol^{-1}

iv) What does your value of K_c suggest about the position of equilibrium in this experiment?

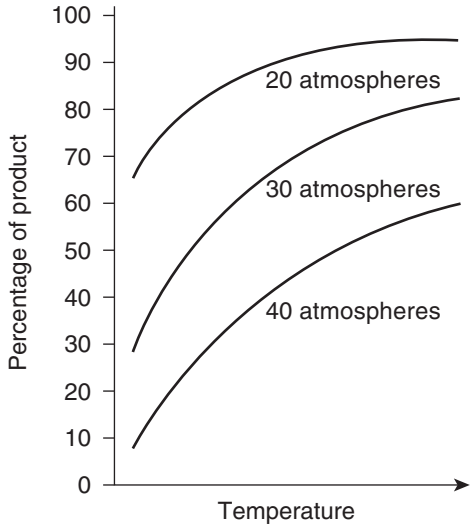
b) The chemist increases both the temperature and the pressure of the equilibrium mixture. The mixture is left to attain equilibrium again.

i) What is the effect, if any, on the value of K_c? Explain your answer.

ii) Explain why it is difficult to predict what would happen to the position of equilibrium after these changes in temperature and pressure.

(OCR Advanced GCE, Chem. A, F325, Jun. 2012, 2)

25 The diagram below shows the effect of temperature and pressure on the percentage of the product in a gaseous equilibrium system.



a) Use the diagram to deduce whether the forward reaction involves an increase or a decrease in the number of moles of gas. Explain your answer.

b) Use the diagram to deduce whether the forward reaction is exothermic or endothermic. Explain your answer.

25 a) The forward reaction involves an increase in the number of moles of gas.
According to the diagram, a higher pressure results in a lower yield. That means the position of equilibrium shifts to the reactant side (the side with a fewer number of moles of gas).

b) The forward reaction is endothermic.
According to the diagram, a higher temperature results in a higher yield. The position of equilibrium shifts to reduce the temperature (to favour the endothermic direction).

26 Many chemical reactions are reversible and are able to form equilibrium mixtures.

Hydrogen and iodine react together to form hydrogen iodide in a reversible reaction.

$$\underset{\text{colourless}}{H_2(g)} + \underset{\text{purple}}{I_2(g)} \rightleftharpoons \underset{\text{colourless}}{2HI(g)} \qquad \Delta H = +53 \text{ kJ mol}^{-1}$$

Refer to page T37.

a) Explain the following observations when changes are made to an equilibrium mixture of $H_2(g)$, $I_2(g)$ and HI(g).

Include reference to the equilibrium position and any other factors.

i) When the temperature is increased, the purple colour becomes paler.

ii) When the pressure is increased, the purple colour becomes deeper.

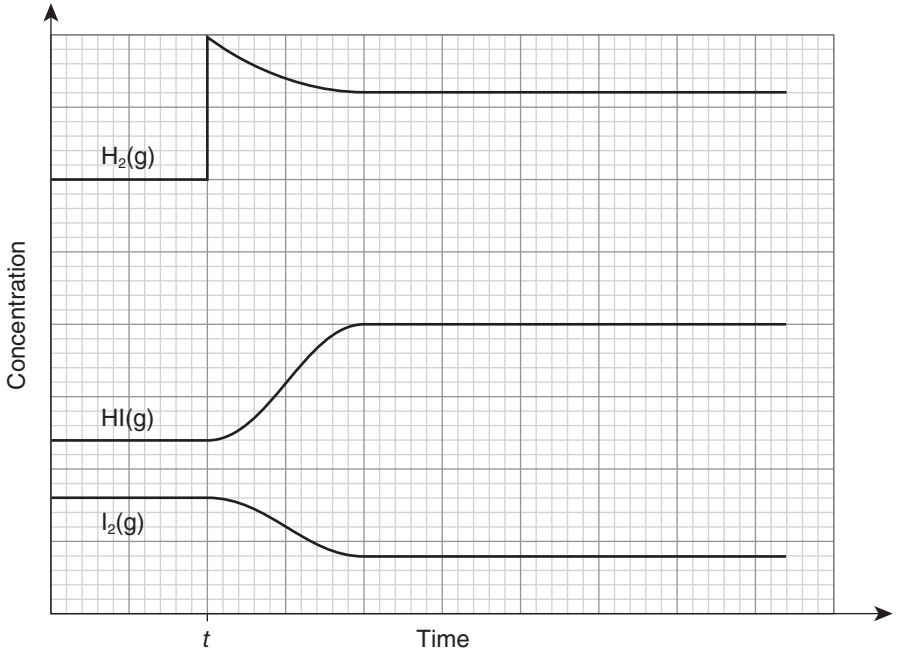
b) The rate at which equilibrium is attained could be increased by increasing the temperature or pressure.

In each case, explain why the rate increases:

i) on increasing the temperature;

ii) on increasing the pressure.

c) $H_2(g)$, $I_2(g)$ HI(g) were mixed together and allowed to attain equilibrium. The concentrations of the gases were then measured at various times and the results plotted. At timed *t*, a change was made to the composition of the mixture.



i) What change was made to the mixture at time *t*?

ii) Explain the changes that happen to the equilibrium mixture after time *t*.

(OCR Advanced Subsidiary GCE, Chemistry, 2813/01, Jun. 2008, 2)